claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_cb1d52fa-fed\agent-ac6475419365f1c23.jsonl`
- SHA-256: `399a7466c3337cb604023c69e019c187ec94def54376ac6774f2ed8bebf51d78`
- Source modified: `2026-06-14T08:29:45+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_cb1d52fa-fed`
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`

Transcript

1. user (2026-06-14T08:27:36.407Z)

THE PIVOT under study = "PERSONALIZATION-FIRST": instead of one universal model for all users, ship a BATTERIES-INCLUDED baseline that CONTINUALLY SELF-OPTIMIZES to a SINGLE user's own footage. As the user's corpus grows (10-15 videos), the on-device model + PER-USER CUE SELECTION (driven by the EXISTING LOVO / A-B experiment harness run on the user's OWN videos, with the user's CORRECTION LOOP as the per-user label source) becomes heavily tuned for THAT user ? deep analysis of THEIR rallies + shot types (serves, smashes, drives, drop shots). Per-user decisions like "the motion/person cue gives no F1 win for THIS user -> default it OFF for them." The model is optimized for the user's videos, NOT others (wins may hold anecdotally on other footage but weaker). Periodic "BATTERIES UPGRADES" improve the shipped baseline so it needs merely <5 videos to reach superior quality. Two user archetypes: (a) an economically-generous hobbyist who records sessions some days; (b) a professional with 100+ hours of their own video. CONTRAST = "generalization-first" (build one model that works on everyone's videos). The owner believes the harness + eval + A/B were DESIGNED to enable this roadmap.

You are a sharp skeptic / contrarian strategist. Here is the parallel research:

[
  {
    "dimension": "PRODUCT / PMF / market",
    "pros": [
      "Sharpest, most ownable value prop in the field: 'your videos, your model, your insights, on your machine.' Every named competitor (SwingVision $179.99/yr, BadPro+ pay-per-use, Dartfish, Hudl) runs a CLOUD + GENERIC-model pipeline. A local model that CONTINUALLY self-tunes to one user's own rallies/shot-mix is genuinely differentiated whitespace none of them occupy today.",
      "Privacy/ownership is a real and growing tailwind, not a marketing fig leaf. 'Images never leave the device' is the core edge-AI selling point and the edge-AI market is large/fast-growing (~USD 25B in 2025, ~22% CAGR). For a personal corpus of your own face/body across many sessions, 'no upload' is a credible, defensible appeal that cloud incumbents structurally cannot match without re-architecting.",
      "Directly weaponizes the existing harness as a PRODUCT engine, not just a dev tool. The LOVO A/B harness (backend/eval/experiment.py compare()/compare_unlabeled), the correction loop as a per-user label source, and per-cue enable flags in IndexingConfig already produce exactly the per-user decision the pivot needs ('motion cue gives no F1 win for THIS user -> default OFF'). The owner's read that these were 'designed to enable this' is substantially correct at the mechanism level.",
      "Sidesteps the project's two hardest commercial blockers at once. Local/on-device eliminates the cloud-GPU hosting unit-economics question (COMMERCIALIZATION.md C7) AND the metered paid-Gemini privacy posture (C4): a self-optimizing local model needs no per-video Gemini call once trained, so there is no recurring marginal cost or 'is your video private' worry.",
      "Pricing model fits the 2026 mood. 'One-time install + batteries-upgrade subscription' rides the documented subscription-fatigue trend (perpetual-license revival, e.g. Capture One $329, Pixelmator $49.99 one-time) while still capturing recurring revenue from periodic baseline upgrades. It is psychologically easier to sell to a hobbyist than yet another $180/yr cloud sub.",

"The 100+hr professional archetype is a strong, underserved fit. A serious player/coach with a large personal corpus is exactly who benefits most from a model tuned to THEIR footage and THEIR shot vocabulary, and is the most likely to (a) already record static full-court and (b) pay a premium for depth that generic tools cannot reach. Personalization quality compounds with corpus size, which aligns price-willingness with value delivered.",
"Quality compounding is a retention/defensibility story per-user: the more you use it, the better YOUR model gets and the higher your switching cost (your tuned model + correction history is sticky and non-portable to a competitor). This is a per-user lock-in that the generic-cloud model lacks."
],
"cons": [
"Cold-start kills the median user. The value only materializes at 10-15 videos (and even the upgraded baseline still needs '<5 videos'). SwingVision/BadPro+ deliver useful output on the FIRST clip. A tool that is mediocre until your 5th-15th session has a brutal time-to-value gap and high early churn before personalization pays off - the opposite of the instant-gratification consumer norm.",
"The 'batteries upgrade' partially cannibalizes the personalization differentiator. If periodic baseline upgrades make it good in <5 videos, the marginal win from per-user tuning shrinks toward the generic baseline - you are simultaneously arguing 'personalization is the moat' and 'the baseline is so good you barely need it.' These two roadmap promises are in tension and must be reconciled in positioning.",
"Per-user overfitting weakens the cross-user data moat that the strategy docs call THE moat ('the moat is the consented dataset + eval harness'). If each model is tuned to one user and wins 'hold anecdotally but weaker' on others, there is no compounding network effect across users - growth does not make the product better for the next buyer, so this is a tool business, not a platform/flywheel business.",
"Contradicts the standing, validated beachhead. PMF_MARKET_RESEARCH.md ranks India badminton ACADEMIES (B2B, buyer=recorder=payer aligned) #1 precisely because individual amateurs are 'low WTP, high CAC/churn.' Personalization-first re-centers on the individual the research deprioritized. The pivot must show why the per-user value prop reverses that low-WTP verdict - it is not self-evident.",
"The static-camera + tripod constraint is unchanged and still gates everything. Personalization does not relax the core capture requirement; it ADDS a corpus-size requirement on top. The addressable user is now 'records static full-court AND has accumulated 10-15 of their own sessions AND owns a capable local machine' - a narrower intersection than the cloud product's.",
"On-device shifts the hardware/support burden to the user and the dev. The repo's own dev container lacks GPU/torch/cv2; a self-optimizing local model implies the END USER has a capable GPU. That excludes most phone-only amateurs, fragments the support surface across consumer hardware, and undercuts the 'just works' install promise - whereas cloud incumbents abstract all of that away.",
"Pricing power is capped by the same free/near-free alternatives the research flagged, and 'continually self-optimizes' is a fuzzy benefit a hobbyist cannot evaluate at purchase time. A one-time fee + upgrade-sub competes against $0 app tiers and BadPro+ pay-per-use; the generous-hobbyist may not perceive 'my model is tuned to me' as worth more than 'good highlights now.'",
"Per-user A/B at n=6-ish videos is statistically thin. The harness's own honest-stats work (bootstrap CI + paired sign test on ?F1) exists because small-n LOVO is not promotable on a bare mean. A single user's corpus is small-n; per-user cue-selection decisions risk being noise-driven ('turn motion OFF for you') and hard to make defensibly without enough of the user's own labels."
],
"findings": [
"Competitive landscape is uniformly CLOUD + GENERIC-MODEL: SwingVision (tennis/pickleball, single iPhone, AI stats/highlights, cloud storage, $14.99/mo or $179.99/yr, 200k+ monthly users), BadPro+ (badminton, pay-per-use), Dartfish & Hudl (coach/team, manual tagging + cloud collaboration, ~$400/yr+). None ship a per-user, self-optimizing, on-device model - the pivot targets genuine whitespace, but whitespace partly because the median consumer wants instant cloud output.",
"Edge/on-device AI is a real, fast-growing trend (USD ~25B 2025 -> ~USD 119B 2033, ~22% CAGR) and its canonical selling point is privacy ('images never leave the device') + low latency. But the documented use cases are surveillance, smart cameras, factories/farms - NOT consumer sports apps, where the incumbents are still cloud. So the privacy angle is credible and on-trend, yet unproven as a CONSUMER sports purchase driver.",
"Pricing reality supports the proposed model: 2025-26 'subscription fatigue' is reviving one-time/perpetual licenses (Capture One $329 perpetual, Pixelmator Pro ~$50 one-time) even as

```
Adobe stays subscription-only. 'One-time install + batteries-upgrade sub' is a defensible, on-
trend hybrid - but consumer software-only sports AI anchors at ~$100-200/yr (SwingVision), which
caps the upgrade-sub ceiling.",
      "The existing PMF research (PMF_MARKET_RESEARCH.md, COMMERCIALIZATION.md) is built on the
OPPOSITE thesis: a closed-source HOSTED API/SaaS, generalization-first, with India badminton
ACADEMIES (B2B) as the #1 beachhead BECAUSE individual amateurs are low-WTP/high-churn.
Personalization-first does not just add a mode - it inverts both the architecture (cloud->local)
and the customer (academy->individual), so the market-of-record must be re-run, not reused.",
      "The mechanism is real and largely built: backend/eval/experiment.py
(compare/compare_unlabeled), calibration.leave_one_video_out (LOVO), per-cue enable flags +
feature persistence (EXPERIMENT_HARNESS.md P2), and the correction loop ->
candidate_segments.human_label give a working substrate for per-user cue selection and per-user
retraining offline, no GPU-cloud needed. The owner's claim that the harness/eval/A-B were
designed to enable this is correct at the engineering layer.",
      "Two archetypes diverge sharply in fit: the 100+hr professional is a strong fit (large
corpus -> deep per-user tuning -> high WTP, already records static) and should be the beachhead
for THIS pivot; the generous-hobbyist is a weaker fit (small/irregular corpus hits the cold-
start wall, lower ability to judge 'tuned to me' value) and is better treated as a funnel than
the primary payer.",
      "The pivot trades a CROSS-user data moat (the strategy docs' stated moat) for a PER-user
switching-cost moat. That is a strategically different business: less platform/flywheel, more
premium-tool-with-lock-in. Whether that is better depends on whether the buyer is the corpus-
rich professional (yes) or the mass hobbyist market (no)."
    ],
    "shapes_future": [
      "Re-pick the beachhead for THIS thesis: lead with the corpus-rich serious player /
individual coach (the 100+hr professional archetype), NOT the academy and NOT the casual
hobbyist. Personalization value scales with corpus size, so target the user who already has the
corpus and the WTP. Re-run the PMF market-of-record under local+individual assumptions before
committing - the current one assumes cloud+B2B.",
      "Solve cold-start as the #1 product problem, since it is the make-or-break gap vs instant-
cloud incumbents. Ship a strong generic 'batteries' baseline that is genuinely useful on video
#1, frame personalization as 'gets even better as it learns YOU,' and instrument time-to-first-
value. The 'batteries upgrade' subscription should be sold as 'fewer videos to greatness' (the
<5-video promise) - make that the headline upgrade benefit.",
      "Make the correction loop the central onboarding UX, not a back-end label source. Per-user
labels ARE the personalization fuel; the product must make correcting/confirming rallies fast,
rewarding, and visibly tied to 'your model just got better.' This doubles as the data-collection
ritual and the retention hook.",
      "Productize per-user A/B honestly: surface 'we tested the motion cue on YOUR videos and it
didn't help, so it's off for you' as a TRUST feature (transparency + 'tuned to you'), but gate
per-user cue toggles behind enough of the user's own labels that the ?F1 sign test / bootstrap
CI clears noise - reuse the existing honest-stats helpers (eval_stats.py) so small-n per-user
decisions aren't noise-driven.",
      "Pricing: anchor a one-time install fee for the batteries baseline + a 'batteries-upgrade'
subscription for periodic model improvements; keep the upgrade-sub well under the ~$180/yr cloud
anchor for the hobbyist, and offer a higher 'pro' tier (deeper per-user analytics, larger
corpora, faster on-device) for the 100+hr professional. Validate with a fake-door test, as the
existing research already prescribes for any pricing.",
      "Treat on-device hardware as a hard product constraint and segment around it: define a
minimum local-GPU spec, offer an optional privacy-preserving local-or-self-hosted path for users
without one, and DON'T promise phone-only mass-market reach. The capture-quality + now corpus-
size + local-hardware triple-gate narrows TAM - own that narrowing as 'serious players only'
positioning rather than fighting it.",
      "Decide explicitly between the two moats. If the goal is a platform/flywheel,
personalization-first undercuts it and a hybrid (per-user tuning ON TOP of a cross-user-
improving baseline, where consented data still feeds the shared 'batteries') is needed to keep
the data flywheel alive. If the goal is a premium per-user tool with lock-in, lean fully into
per-user and drop the cross-user-moat framing from the strategy docs to avoid relitigating it."
    ],
    "open_questions": [
      "Will a corpus-rich serious player/coach actually pay MORE for 'tuned to my footage' than
for SwingVision/BadPro+'s instant generic output - or is per-user depth a 'nice but not worth
it' benefit they can't evaluate at purchase? Needs the fake-door + 5-10 buyer interviews the
research already prescribes, re-targeted at individuals not academies.",
```

"How big is the cold-start chasm in practice - what is rally-detection F1 on a user's video #1 with only the generic baseline vs at video #10-15 with per-user tuning, and is the video-#1 quality already competitive with cloud incumbents? This single benchmark gates whether the median user ever reaches the personalization payoff.",
"Does 'continually self-optimizes to ME' produce a quality delta a non-expert hobbyist can SEE and value, or only a metric a coach/pro cares about? If the gain is invisible to the buyer, the differentiator doesn't convert to WTP.",
"What local hardware does the self-optimizing model actually require, and what fraction of the target users own it? This sets the real TAM after the static-camera + corpus-size + local-GPU triple gate, and decides whether a self-hosted fallback is mandatory.",
"Is the per-user switching-cost moat worth giving up the cross-user data flywheel the strategy docs call THE moat - i.e. is this a premium-tool business or a platform business? The owner must choose, because the docs currently assume the platform thesis the pivot undercuts.",
"Can per-user cue-selection decisions (e.g. 'motion OFF for you') be made defensibly at one user's small-n corpus without the harness's honest-stats gating flagging them as noise - and how many of the user's OWN corrected labels are needed before a per-user toggle is statistically earned?",
"How are 'batteries upgrades' delivered and licensed for an on-device model (ship new weights? local fine-tune recipe?), and does that distribution re-open the commercial-clean weights-provenance question (C3) and any open-weights distribution obligations the cloud-only posture currently avoids?",
"Does the privacy/ownership appeal that sells in surveillance/enterprise edge-AI actually move a CONSUMER sports buyer's purchase decision, or is it a tie-breaker at best? No evidence yet that 'your videos never leave your machine' is a primary driver for amateur athletes."
]
},
{
"dimension": "Technical feasibility of per-user continual personalization (personalization-first pivot)",
"pros": [
"TWO-LAYER SPLIT ALREADY EXISTS AND IS THE EXACT ENABLER. The repo's shipped 'separate expensive DETECTION from cheap DECISION' thesis (docs/EXPERIMENT_HARNESS.md sec.2; backend/eval/experiment.py::compare) means detection runs once per video and is persisted (candidate_segments.stats JSON), then ANY number of decision variants re-score offline, deterministically, with no GPU. Per-user personalization == per-user re-scoring of THAT user's persisted candidates. The owner's belief that the harness was 'designed to enable this' is structurally correct at the decision layer: a Variant (experiment.py:76) is a JSON recipe, and per-user tuning is choosing the variant/thresholds that win on the user's own videos.",
"THE PERSONALIZABLE ARTIFACTS ARE GENUINELY TINY AND CHEAP. The learned arbiter is backend/eval/classifier.py::LogisticRegression: pure numpy, ~5 features, L2, fits in milliseconds, serializes to a kilobyte JSON (to_dict/save). The Gen-0 TAL head (rally_seq_proto.py / ASFormer) is ~1.1M params, trains in minutes at ~4.5GB VRAM for $0 on the owner's 2070 Super (OWNED_MODEL_IMPLEMENTATION_PLAN.md M0.4). A per-user model = one small JSON/checkpoint + a per-user (velocity_thresh, merge_gap, min_window_duration, min_crossings) tuple. Per-user retrain on a hobbyist machine is feasible-now for THIS class of model.",
"PER-USER CUE SELECTION IS ALREADY A FIRST-CLASS, MEASURED OPERATION. The pivot's 'motion/person cue gives no F1 win for THIS user -> default OFF' is exactly compare(videos, control, fusion) + the honest gate (eval_stats.paired_delta_ci: CI-excludes-zero AND sign-test agree). This is not hypothetical: the first end-to-end A/B (2026-06-first-ab-experiment) ran it on n=6 and produced a clean YES (Gate-A net_crossings>=2: +0.074 F1, 6/6 wins, p=0.031) and a clean NO (the window-level logistic: -0.081). The machinery to make a per-user cue ON/OFF decision honestly already runs offline.",
"PER-VIDEO / PER-USER CALIBRATION IS THE REPO'S OWN STATED PRODUCT IDEA, EMPIRICALLY MOTIVATED. HOW_RALLY_DETECTION_WORKS.md explicitly names 'per-video calibration as a feature' ('watch a few labeled rallies -> auto-tune to this court/lighting -> segment the rest'). The multivideo-LOVO study found the cross-video calibration gap (not representation) is the n=2 wall: Kushagra's learned held-out F1 went 0.049 -> 0.417 -> 0.561 as calibration improved, and probability-scale shift across footage is the documented failure. A single user's corpus is far more homogeneous (same court, camera, lighting, players) than the cross-video corpus, so within-user calibration should be EASIER and more stable than the general cross-video problem the repo already partly solves.",
"CATASTROPHIC FORGETTING IS LARGELY DESIGNED-AROUND BY ARCHITECTURE, NOT A NEW RISK. The shuttle substrate (WASB) is FROZEN by design (HOW_RALLY_DETECTION_WORKS.md). Personalization touches only the lightweight decision head + thresholds; the perception backbone never updates per user, so the classic on-device forgetting failure mode (overwriting backbone knowledge) does

not apply to the cheap layer. Web-grounded: frozen-backbone + lightweight-adapter
personalization (PROPER per-user LoRA, P2P hypernetwork-from-profile, ~300-param structure-gated
biases) is an established 2025 pattern precisely because freezing the backbone 'substantially
mitigates forgetting'. The repo's frozen-WASB + tiny-head design is already in this safe
regime.",
      "COLD-START HAS A REAL, ALREADY-SCOPED FALLBACK. A brand-new user with <5 videos falls
back to the shipped baseline: the pinned CONTROL variant (experiment.py:158) + the label-free
decision-level gate (MULTI_SIGNAL_FUSION_PLAN.md sec.3.1: Gate A net-crossings AND Gate B court-
occupancy need NO labels). Gate-A is the validated, promote-worthy, zero-label win that ships
before any per-user labels exist. So 'batteries-included baseline' on day 0 is not aspirational;
the label-free fusion gate is the concrete day-0 product, and Gemini remains the always-ready
inference fallback (never a training source).",
      "THE CORRECTION-LOOP SUBSTRATE PARTIALLY EXISTS. candidate_segments carries human_label +
confirmed_segment_id (database.py:109-110); query_candidate_segments can filter only_unlabeled.
So the DB seam for 'user corrections become per-user labels' is reserved, and
training.label_candidates already turns intervals into y by the evaluator's own IoU matcher (the
same matcher used for scoring) -- meaning user-confirmed segments could feed the per-user
classifier through existing, tested glue."
    ],
    "cons": [
      "THE SMALL-N STATISTICS PROBLEM IS THE CENTRAL, UNRESOLVED RISK -- AND IT GETS WORSE PER-
USER, NOT BETTER. The whole eval_stats.py module exists BECAUSE a single LOVO mean lies at n=6
(its docstring: 'leave-one-out anti-correlates each fold's training and held-out label means at
small N, and one lucky/unlucky fold swings the headline'). Per-user A/B/LOVO runs on a SINGLE
user's tiny corpus, so a 10-15 video hobbyist is at n=10-15 max, and a fresh user is at n=1-4
where LOVO is undefined (leave_one_video_out requires >=2 videos and returns an error otherwise;
calibration.py:122). The pivot proposes making per-user ON/OFF cue decisions exactly where the
repo's own honesty tooling says a bare mean is untrustworthy. Web-grounded: 'numerical cross-
validation scores turn out not to be useful for small data sets, as the variance is too large to
make any robust statements.' Risk: per-user cue selection will frequently fire on noise (turn a
cue OFF for a user when it actually helps, or vice versa).",
      "THE FIRST REAL A/B ALREADY SHOWED THE SMALL-N FAILURE MODE CONCRETELY. In 2026-06-first-
ab-experiment, the window-level learned variant scored F1=0.000 on testlarge_short (6 rallies)
-- it predicted ZERO rallies, a pure threshold/calibration artifact at small N. A per-user
system on a hobbyist with one short session would hit this same cliff: the learned head can
collapse to all-negative when the operating point doesn't transfer. The repo's own fix
(tune_threshold + calibrate inside the LOVO loop; experiment.py:265, _calibrate_and_tune) is
built but UNVALIDATED, and tuning the threshold on a tiny train fold is itself a small-N
overfitting source (web: 'aggressive hyperparameter tuning leads to overfitting... the model
becomes overly specialized... focusing on noise').",
      "ON-DEVICE FINE-TUNING OF THE HEAVYWEIGHT VLM (Qwen3-VL) IS LIKELY INFEASIBLE ON A
HOBBYIST MACHINE -- AND EVEN INFERENCE IS UNMEASURED. OWNED_MODEL_IMPLEMENTATION_PLAN.md is
explicit: Qwen3-VL fine-tune needs rented 24-48GB A100/L40S (ms-swift multimodal-GRPO examples
use 6-8 GPUs; 'budget is the largest open risk and is currently unquantified'); the 2070 Super
'CANNOT QLoRA a 7B video VLM (~20-24GB)'. Even on-device 4B INFERENCE latency on 8GB Turing at
1080p/60fps is listed as an explicit unmeasured open risk. Web-grounded: QLoRA brings a 7B to
~8-10GB and a 10B to ~11.6GB -- a floor, before video tokens (which 'can balloon'). So per-user
VLM personalization on a hobbyist's machine is NOT feasible-now; it needs cloud bursts (re-
introducing the DPA/consent/no-train-on-minors machinery the local-first pivot was meant to
avoid) or is restricted to the professional archetype with serious hardware.",
      "'OPTIMIZED FOR THIS USER, WEAKER ON OTHERS' IS A DELIBERATE OVERFIT, AND THE REPO HAS NO
PER-USER REGRESSION GUARD. The no-regression gate (run_regression.py + regression_baseline.json)
is built around a SHARED golden baseline -- 'rollback = flip a config flag.' A per-user model
that is intentionally tuned to one user has no shared baseline to regress against; the system
would need a PER-USER baseline + per-user no-regression discipline that does not exist. Without
it, a 'batteries upgrade' or a per-user retrain can silently degrade a user's experience with no
automated catch, which directly contradicts the owner's load-bearing 'no regression, ever'
requirement.",
      "CATASTROPHIC FORGETTING REAPPEARS AT THE 'BATTERIES UPGRADE' SEAM IF THE BASELINE IS A
FINE-TUNED VLM. The cheap layer is forgetting-safe, but the pivot's '<5 videos to superior
quality' implies the SHIPPED baseline itself keeps improving via fine-tuning across users'
consented data. Continually fine-tuning a shared base model across batteries-generations is the
exact regime the 2025 surveys flag: 'PEFT techniques still lack the ability to adjust models to
multiple tasks continually due to catastrophic forgetting.' Each batteries upgrade risks
regressing the cohorts the previous baseline served well -- needing replay buffers / parameter-

isolation / careful eval, none of which is designed yet. This is needs-research, not feasible-now.",
      "THE PER-USER LABEL SOURCE (CORRECTION LOOP) IS A SEAM, NOT A PIPELINE. human_label/confirmed_segment_id columns exist, but there is NO wired flow from 'user edits a rally boundary in the UI' -> 'per-user label row' -> 'per-user classifier retrain' -> 'per-user variant promotion.' The labels that actually feed distill_local today are Gemini SILVER labels, which the guardrails forbid as a TRAINING source for owned weights (OWNED_MODEL M0.3(i): distill_local.py is the live no-paid-LLM-training violation that must be purged). So the per-user training-label supply chain has to be built from scratch on the human-correction side; it cannot reuse the existing Gemini-silver path.",
      "PER-USER COMPOUNDS LABEL SCARCITY: corrections are sparse, biased, and noisy. A hobbyist who 'records some days' generates very few corrections, and users correct what's visibly wrong (false positives they notice) far more than what's silently missed (false negatives) -- a labeling bias that systematically skews a per-user classifier toward precision at recall's expense. The multi-court 'proposal-recall ceiling' finding (kushagra 0.125, gbaaddy 0.312) shows the decision layer CANNOT recover rallies the windowing missed; per-user decision tuning cannot fix a per-user perception/proposal failure, so personalization has a hard ceiling set by the frozen substrate on that user's footage.",
      "COLD-START QUALITY CLAIM ('<5 videos to superior quality') IS CURRENTLY UNSUPPORTED BY THE DATA. The honest cross-video heuristic bar is ~0.480 LOVO F1 and the learned prototype ~0.583 at n=6 -- neither is 'very high P/R,' and the ship-gate target for 'very high' is explicitly UNDEFINED (OWNED_MODEL open risks). There is no evidence yet that <5 of a NEW user's videos reach 'superior' quality; the repo's own data says quality is gated by data scale + multi-court isolation, and a new user's first videos may well be the hard multi-court academy case (the most market-representative one) where even the learned model is windowing-capped."
    ],
    "findings": [
      "The 'detection once, decision re-scored offline forever' architecture (docs/EXPERIMENT_HARNESS.md sec.2; backend/eval/experiment.py) is REAL and shipped, and it is the genuine technical enabler of cheap per-user personalization AT THE DECISION LAYER. The owner's intuition that the harness enables this roadmap is correct for the lightweight layer (thresholds + the numpy LogisticRegression + per-cue ON/OFF), and wrong/unproven for the heavyweight layer (per-user VLM fine-tuning).",
      "What personalizes cheaply, feasible-NOW on a hobbyist machine: (a) the 4 segmentation knobs (velocity_thresh, merge_gap, min_window_duration, min_crossings -- calibrate_local.py LocalConfig), (b) the ~5-feature LogisticRegression arbiter (classifier.py, kilobyte JSON, ms training), (c) per-user cue ON/OFF via compare()+paired_delta_ci, (d) optionally the ~1.1M-param Gen-0 TAL head ($0, minutes, 4.5GB on a 2070 Super). What does NOT personalize cheaply: the Qwen3-VL VLM -- on-device fine-tune needs 24-48GB rented cloud (explicitly 'cannot QLoRA on the 8GB box'), and even quantized on-device INFERENCE on 8GB is an unmeasured open risk.",
      "The small-N statistics problem is the dominant feasibility risk and it is STRICTLY WORSE per-user than the n=6 corpus problem the repo already grapples with. The repo BUILT backend/eval/eval_stats.py (bootstrap CI + paired sign test) specifically because a bare LOVO mean lies at small N; per-user runs at n=1-15 where LOVO is undefined-to-noisy. The first real A/B already exhibited the failure (learned variant -> F1 0.000 on the 6-rally video via a threshold/calibration artifact). Web-grounded: CV scores are 'not useful for small data sets -- the variance is too large to make robust statements.'",
      "Catastrophic forgetting is bifurcated: SAFE for the cheap per-user layer (frozen WASB backbone + tiny head is the established forgetting-resistant frozen-backbone-adapter pattern), but RE-EMERGES at the 'batteries upgrade' seam if the shipped baseline itself is a continually-fine-tuned VLM across cohorts -- the exact regime 2025 surveys flag as unsolved for PEFT. Designed-around for v1 (cheap layer), needs-research for the upgrade cadence (heavy layer).",
      "The correction-loop label pipeline and a per-user no-regression guard are the two missing engineering seams. The DB columns (human_label, confirmed_segment_id) and the IoU label-maker (training.label_candidates) exist, but the wired flow correction->per-user-label->retrain->per-user-promotion does not, and the existing no-regression gate assumes a SHARED golden baseline that a deliberately-overfit per-user model has no analogue for. Also: the only currently-wired training-label source (Gemini silver, distill_local.py) is licensing-forbidden as a training source and must be replaced by the human-correction supply on this path."
    ],
    "shapes_future": [
      "DECOUPLE THE TWO PERSONALIZATION LAYERS EXPLICITLY in the roadmap: (1) cheap per-user DECISION personalization (thresholds + LogisticRegression + cue ON/OFF + Gen-0 head) is feasible-now and should be the v1 of the pivot; (2) per-user/heavy VLM personalization is needs-research and cloud-bound and must stay gated behind the same Phase-2 budget benchmarks already

flagged. Do NOT let the pivot's framing collapse these -- the owner's 'continually self-optimizes' must mean the cheap layer for v1.",
      "ADOPT A PER-USER PROMOTION GATE that requires the honest signal, not a bare mean: a per-user cue/variant flips ON only when paired_delta_ci.ci_excludes_zero AND the sign test agrees on THAT user's held-out videos, with an explicit minimum-N floor (e.g. refuse per-user learned promotion below n>=5 user videos and fall back to the shipped baseline + label-free Gate-A/Gate-B). The first-A/B's F1=0.000 artifact is the concrete reason this guard is mandatory before any per-user auto-tuning.",
      "BUILD THE COLD-START AS THE LABEL-FREE FUSION GATE (Gate-A net-crossings>=2, validated +0.074/6-of-6/p=0.031, plus Gate-B court occupancy), since it needs zero per-user labels and is the only currently-validated 'batteries' win -- then layer per-user learned tuning on top only as the user's corpus and corrections accumulate. This makes '<5 videos -> good baseline' concrete (label-free gate) and defers the unproven '<5 videos -> superior personalized' claim to a measured milestone.",
      "WIRE THE CORRECTION LOOP AS THE PER-USER LABEL SOURCE on the human side (human_label/confirmed_segment_id -> per-user golden rows -> training.label_candidates), explicitly REPLACING the licensing-forbidden Gemini-silver training path (distill_local.py / OWNED_MODEL M0.3(i)), and instrument it for the correction-bias risk (users report false positives far more than misses -> add a recall-protecting prior or active-learning prompts for likely-missed rallies).",
      "TREAT 'BATTERIES UPGRADE' AS A CONTINUAL-LEARNING PROBLEM WITH A REGRESSION HARNESS PER COHORT: before shipping a new baseline that needs '<5 videos,' run the no-regression gate against a held-out multi-user/multi-court eval set (including the windowing-capped multi-court matches) to catch cross-cohort forgetting. The professional archetype (100+ hrs, real hardware) is the right early adopter for any VLM-level personalization; the hobbyist archetype should be served by the cheap-layer pivot only until on-device VLM inference latency on 8GB is actually MEASURED (it is currently an open risk)."
    ],
    "open_questions": [
      "What is the minimum per-user corpus size N at which per-user cue ON/OFF selection (paired_delta_ci on the user's own LOVO) beats simply using the shipped label-free Gate-A/Gate-B baseline? The repo proved learned-beats-heuristic at n=3 on a DIVERSE cross-video corpus -- but a single user's HOMOGENEOUS corpus has lower variance (could converge faster) yet far fewer independent folds (could never reach significance). This is empirically resolvable on the GPU/WSL box by re-running ab_n6_lovo.py restricted to within-user splits, and is the single highest-value experiment for the pivot.",
      "Does within-user calibration actually transfer better than cross-video (the pivot's core bet)? The multivideo-LOVO study showed cross-video probability-scale shift is the n=2 wall; a same-court/same-camera user corpus should NOT shift -- but this is asserted, not measured. Need a 2+ video single-user holdout to confirm the learned operating point transfers across that user's own sessions without per-video re-calibration.",
      "Can a hobbyist's machine run the SHIPPED baseline (frozen WASB-in-WSL + torchvision person cue + the tiny head) at acceptable latency, and is per-user retrain of just the head/thresholds fast enough to feel 'continual'? WASB runs in WSL on the owner's 2070 Super today; the open question is the lower-bound consumer GPU (8GB or less, or CPU-only) on which the cheap layer is still usable -- unmeasured.",
      "For the professional archetype (100+ hrs), is a per-user QLoRA of the Gen-2 VLM worth it vs the cheap layer, and does it hold under the catastrophic-forgetting + cloud-DPA constraints? This needs the still-unquantified VLM VRAM/token/$-per-call benchmark on real 1080p/60fps clips (OWNED_MODEL open risk #2/#4) before it can even be scoped.",
      "How is per-user data deleted and a per-user model 'unlearned' on request (GDPR), given the pivot stores per-user adapters/labels? The plan says 'deletion = drop the adapter' for per-user adapters, but a baseline that was 'batteries-upgraded' on consented cohort data cannot un-bake one user without retraining -- the unlearning story for the shared baseline is undefined.",
      "Does the correction-loop label bias (users flag false positives >> false negatives) systematically push per-user models toward over-suppression, and what recall-protecting mechanism (active-learning prompts for probable-missed rallies, asymmetric class weighting) is needed? Untested -- and directly threatens the multi-court footage where the proposal-recall ceiling is already the binding constraint."
    ]
  },
  {
    "dimension": "How the existing harness/architecture supports personalization-first (verifying the owner's \"designed for this\" claim)",
    "pros": [

"LOVO/compare() is corpus-relative by construction, so 'a per-user corpus' is literally just the list of `VideoCandidates` you pass in. `compare(videos, control, fusion_person_off)` over ONE user's videos already answers 'does the person cue win for THIS user?' with zero new code (backend/eval/experiment.py:403). The owner's claim that 'motion cue off for this user IS a per-user Variant decision' is TRUE at the API level.",

"Per-user cue gating is already a first-class Variant knob. `Variant.cue_flags`, `min_crossings` (Gate A), `min_players` (Gate B / person cue) and `feature_names` are exactly the per-user switches the pivot needs; setting `min_players=0` / dropping the person feature column = 'person cue OFF for this user' (experiment.py:108-125, 200-210). Promotion-only-on-lift (`compare` -> ?F1 CI clears 0 AND sign test agrees) is precisely 'turn a cue ON for a user only if it helps that user' (eval_stats.py:90-109).",

"The replayable feature substrate is real and is the strongest support for 'continually self-optimize as the corpus grows.' fusion_golden.py writes ONE small JSON per video (`<name>_golden_features.json`) holding shuttle+person features per candidate window; the GPU touches each video exactly once, and re-scoring any variant later is pure-CPU offline (fusion_golden.py:1-16, 123-153). Adding the user's 11th video = extract its features once, then re-run LOVO over the whole grown corpus. This 'detect once, decide forever' split (EXPERIMENT_HARNESS.md §2) is exactly the per-user-corpus-growth loop.",

"Features are scale/translation-invariant and default-OFF (fusion_features.py:1-21, 192-197), which is what lets a cue be carried in production but only ACTIVATED per-user. `FusionSegmenter` already persists `net_crossings` + the 5 person features into production `candidate_segments.stats` and gates them via `indexing.fusion.min_crossings/min_players` config (fusion_hybrid.py:76-151) ? a real per-deployment knob, and config is per-deployment today, so single-user-per-deployment (the hobbyist) gets per-user cue selection for free.",

"The honest small-N statistics (C1?C4) are *more* valuable for personalization than for the generic model, because a single user's corpus is exactly the n?6 regime eval_stats.py was built for: bootstrap CI + exact paired sign test (eval_stats.py:69-109), per-video CIs and F1@k + segment-count-ratio (experiment.py:359-400). A per-user promotion decision off n=10 videos without these would be noise; the harness already refuses to promote on a bare mean.",

"The per-corpus model-fitting primitive already exists. `distill_local.run()` fits a 'final model on ALL videos, for running on a fresh held-out video' (distill_local.py:132-142) ? i.e. 'train on this user's corpus, apply to their next video.' `LogisticRegression.save/load` makes that fitted model a single pinnable JSON artifact (the natural per-user model file). The math for 'a model tuned to THIS corpus' is done.",

"The leaderboard (`record_experiment`) already stamps each result with a `label_set_hash`, variant `spec_hash`, `num_videos` and the honest ?F1 (experiment.py:504-540). One JSON row per corpus-vs-variant is a natural per-user experiment log; keying it by user is an additive field, not a redesign ? consistent with the owner's intent."

],

"cons": [

"The single biggest gap: the learned per-user model has NO path to production serving. `classifier_model` / `LogisticRegression.load` is referenced ONLY in backend/eval/* ? grep of backend/pipeline finds zero loads (the served `FusionSegmenter` uses only the `min_crossings`/`min_players` gates, fusion_hybrid.py:133-151). So 'optimize a per-user model' currently produces an artifact the live pipeline cannot consume. Per-user TUNING of gate thresholds is supported; per-user learned MODELS are not served. This materially weakens 'the model continually self-optimizes to the user.'",

"The 'correction loop -> per-user labels' pipeline does NOT exist as a write path. The schema has `candidate_segments.human_label` and `link_candidate_to_segment` (database.py:110, 274-286), and the pipeline auto-links confirmed AI segments back to candidates (yolo_hybrid.py:348-355), but NOTHING in production ever SETS `human_label` ? the only writer is a test doing raw SQL (test_database.py:152). There is no correction/reject/approve API endpoint in main.py. The 'user's corrections as the per-user label source' is a schema stub, not a working loop ? the most load-bearing assumption of the pivot is the least built.",

"Everything is single-corpus / single-tenant by design and explicitly scoped that way. There is NO user/tenant dimension anywhere (`user_id`/`tenant` appear only as a *future* path-containment comment, config/models.py:207). The harness A/B's 'a corpus' but has no notion of WHICH user's corpus, no per-user model store, no per-user model versioning, no per-user eval report. EXPERIMENT_HARNESS.md §9 pins it as 'single-tenant/local-first ... explicitly NOT MLflow/W&B/a service' ? i.e. the harness was designed for ONE corpus (the owner's golden set), and 'per-user' is a multiplication of that it was never built to manage.",

"Today the labels driving LOVO are GOLDEN (vendor-produced, IoU-matched) or Gemini-silver ? NOT the user's own corrections. fusion_golden.py loads labels from `*.rallies.csv` golden files (fusion_golden.py:28-46, 62-64). Swapping in noisy per-user click-corrections as `gts` changes the statistical regime the C1?C4 honesty was validated for, and there is no de-biasing /

partial-label handling for 'the user only fixed the rallies they noticed.'",
      "No on-device / incremental training loop exists. `LogisticRegression.fit` is a full batch refit over the pooled corpus (distill_local.py:116, 133-135); there is no incremental update, no scheduled re-fit trigger as videos arrive, no 'periodic batteries upgrade' mechanism. 'Continually self-optimizes' is currently 'an operator can re-run a CLI'; the continual/automatic part is unbuilt.",
      "The pivot's claim that wins need only hold for THIS user (and may be weaker elsewhere) is in tension with the repo's stated training discipline: OWNED_MODEL_IMPLEMENTATION_PLAN frames ONE shared model across sports/users and requires 'split data by match' + 'per-user training data needs a separate explicit opt-in' (NEXT_STEPS context lines 29-31). The harness enforces cross-VIDEO honesty (LOVO), but a per-user model that overfits a user's court/camera is exactly the 'memorise this court' failure fusion_features.py:1-9 warns against ? the per-user framing partially fights the invariance discipline the features were built around.",
      "The owner's 'designed for this' is retrospective fit, not original intent. Every doc (EXPERIMENT_HARNESS.md, MULTI_SIGNAL_FUSION_PLAN.md) frames the harness as 'add cues without regressions to the SHIPPED baseline,' a generalization-first goal. The same primitives HAPPEN to compose into per-user optimization (genuinely ? see pros), but there is no design doc, test, or code comment that names personalization/per-user as a target. The claim is 'the bones support it,' which is true; 'it was designed for it' overstates the record."
    ],
    "findings": [
      "VERDICT on the owner's claim: PARTIALLY TRUE, and impressively so for the DECISION layer, but overstated for the end-to-end loop. The harness's corpus-relative LOVO + Variant-as-recipe + promote-only-on-lift + replayable feature substrate genuinely make 'per-user cue selection' a special case of what already runs ? you can run it on one user's videos today with no new code. But three of the pivot's load-bearing pieces are stubs or absent: (1) per-user LEARNED models can't be served, (2) the correction-loop->labels write path doesn't exist, (3) there is no per-user/tenant dimension, model store, or eval report.",
      "ALREADY BUILT (reusable as-is for per-user): `compare()`/`evaluate_variant()` LOVO over an arbitrary video list (experiment.py:296-448); `Variant` with per-cue flags + Gate A/B thresholds + pinnable classifier path (experiment.py:76-153); honest small-N stats C1?C4 ? the n?6 regime IS the per-user regime (eval_stats.py; segmentation_metrics.py); replayable per-video feature JSONs decoupling GPU detect from CPU decide (fusion_golden.py); production persistence of cue features into `candidate_segments.stats` with served Gate A/B config knobs (fusion_hybrid.py); per-corpus batch model fit + save/load (distill_local.py:132-142, classifier.py); the leaderboard with label-set + spec hashes (experiment.py:504-540).",
      "MISSING / STUBBED (the gap list the pivot must fund): (a) serving path for a learned per-user model ? `LogisticRegression.load` is never called in backend/pipeline; only the gates are served (fusion_hybrid.py:133-151). (b) correction-loop write path ? `human_label` is never SET in production; no correction/reject API in main.py (only the schema column + linkage exist, database.py:110/274-286). (c) per-user identity & storage ? no user_id/tenant, no per-user model store, no per-user model versioning, no per-user leaderboard partition (single-corpus by design, EXPERIMENT_HARNESS.md §9). (d) on-device / continual / scheduled re-fit ? only a manual CLI batch refit (distill_local.py). (e) per-user label handling ? LOVO today consumes golden/silver labels, not noisy partial user corrections; no partial-label de-biasing.",
      "ARCHITECTURAL ASYMMETRY worth flagging to the owner: the harness supports per-user TUNING (pick gate thresholds / which cues to enable for this corpus) far better than per-user LEARNING (a model fit to this user's footage), because tuning outputs config the served `FusionSegmenter` already reads, whereas learning outputs a JSON model nothing in production loads. If the pivot leans on 'deep analysis of THEIR shot types' via a learned model, that serving gap is the first thing to close; if it leans on per-user cue ON/OFF + threshold selection, the harness is nearly ready."
    ],
    "shapes_future": [
      "The cheapest credible first slice of personalization-first is per-user CUE SELECTION via the EXISTING harness: run `compare(this_user_corpus, control, person_off_variant)` and write the winning `min_crossings`/`min_players`/`cue_flags` into that deployment's `indexing.fusion.*` config. No new ML, no serving change ? it rides fusion_hybrid.py's existing knobs. This validates the pivot's core promise ('person cue off for THIS user') with days of work, and should precede any per-user-model investment.",
      "To serve per-user LEARNED models, the missing seam is small but real: teach `FusionSegmenter._select_for_handoff` (fusion_hybrid.py:133-151) to optionally load a pinned `LogisticRegression` and score the persisted feature columns ? the eval-side `predict()` (experiment.py:213-240) is the exact reference implementation to mirror. This single bridge is the highest-leverage unlock and is currently the wall between the harness and the pivot.",

"The correction loop must become a real write path before 'user corrections as labels' is more than aspiration: a `set_human_label`/confirm-reject API in main.py + a setter on the DB (the column and `link_candidate_to_segment` already exist). Until then, per-user LOVO can only run on golden/Gemini labels, so the demo of personalization will quietly be 'tuned to vendor labels on the user's videos,' not 'tuned to the user's own corrections' ? an honesty risk given the repo's attribution discipline.",
"Per-user scale forces decisions the single-tenant harness deferred (EXPERIMENT_HARNESS.md §9): a per-user model store + versioning, a per-user leaderboard partition (add a `user_id`/corpus key to the `record_experiment` row ? additive), and a re-fit trigger as the corpus grows. These are the difference between 'an operator re-runs a CLI' and the pivot's 'continually self-optimizes'; they're additive to the current design but are NOT in it.",
"The invariance discipline (fusion_features.py:1-9: features are counts/ratios/zones, never raw pixels, to avoid memorising one court/camera) is a double-edged asset for personalization. It's GOOD (a per-user model still generalizes across the user's sessions/venues) but it CAPS how 'deeply tuned to THEIR court' a model can get ? the pivot's 'wins may not transfer to other users' upside is partly engineered away by design. If the owner wants stronger per-user specialization, that's a deliberate feature-design departure to make consciously, not by accident."
],
"open_questions": [
"What is the per-user LABEL source in practice ? the user's accept/reject of highlight clips, or boundary edits? The harness needs IoU-matchable interval GTs (label_candidates, training.py); thumbs-up/down on a reel is a weaker signal that needs a defined mapping to `gts` before per-user LOVO is meaningful.",
"For the two archetypes, does 'per-user model' mean a per-user LEARNED classifier (needs the serving bridge + correction loop) or per-user CONFIG (cue flags + thresholds, already serveable)? The hobbyist (10-15 videos) realistically only supports the latter honestly at n?10; the 100h-pro could support a learned model. The pivot may need two different mechanisms per archetype.",
"How does a 'periodic batteries upgrade' to the shipped baseline interact with a user's accumulated per-user tuning ? does the upgrade reset their cue-selection, or must per-user deltas be re-derived against the new baseline each upgrade? The leaderboard's `spec_hash`/`label_set_hash` can detect staleness but there's no re-derivation policy.",
"Is per-user data ever pooled to improve the shipped baseline? The guardrails (NEXT_STEPS: 'per-user training data needs a separate explicit opt-in', exclude minors) and the paid-LLM-not-a-training-source rule constrain this; the pivot's 'batteries upgrades' need a defined, opt-in, minor-excluded pooling path that today does not exist.",
"Given features are deliberately court/camera-invariant, how much measurable per-user F1 lift is actually achievable beyond cue ON/OFF selection? Worth an early ablation on a single user's corpus (one of the existing golden videos as a proxy) to size the upside before funding the per-user serving + storage build-out."
]
},
{
"dimension": "DATA / LABELS / PRIVACY / LICENSING",
"pros": [
"Cleanest licensing posture the project can have: a user's own footage is commercial-CLEAN by construction (no third-party copyright, no scraped corpus, no ToS to launder). It sidesteps the project's biggest unresolved legal risks at once -- it never needs paid-LLM outputs as a training source (CLAUDE.md hard rule; the live violator backend/eval/distill_local.py training on Gemini-silver becomes moot for the per-user path), and the per-user TAL/cue weights inherit no provenance debt the way scraped academic data does.",
"On-device personalization is the strongest privacy story available and is already the documented end-state, not a new invention: STORAGE_SHARING_MODEL.md L131-135 says the heaviest users run the pipeline locally and sync only reels+metadata; OWNED_MODEL_IMPLEMENTATION_PLAN.md frames on-device as the deferred compression goal. 'Data never leaves the machine' makes GDPR/DPDP largely N/A for the per-user training loop (no DPA, no cross-border transfer, no cloud retention) -- a real differentiator vs Gemini, which 'can never ship on-device' (the licensing guardrail literally becomes the moat).",
"The correction loop as the per-user label source is elegant AND already half-built, not speculative: the schema substrate exists today -- candidate_segments.human_label and confirmed_segment_id columns (database.py:109-110), query_candidate_segments(only_unlabeled=True) (the pseudo-label-then-correct flow), and label_candidates() (training.py, a pure source-agnostic y-labeler). 'Pseudo-label then correct' is exactly what the table was shaped for, and v1 scope (CLAUDE.md: 'rally + highlight + history

```
+ correction loop') already commits to it. The harness consumes these labels with zero new
scoring code (EXPERIMENT_HARNESS.md: compare + LOVO + record_experiment all shipped
2026-06-13).",
      "Per-user labels are higher-signal-per-unit than a generic golden set for THAT user: the
labels come from the exact camera, court, lighting, and players the model will serve,
eliminating the corpus-representativeness gap that GOLDEN_SET_VENDORING.md L34/Q2 explicitly
flags as 'strong evidence, not a guarantee' for a licensed/owned corpus. Personalization turns
that honest caveat into a non-issue per user -- the eval corpus IS the deployment corpus.",
      "Roora's role SHARPENS rather than shrinks: it stops being a per-user golden oracle (which
never scaled -- you cannot vendor every user's private footage; GOLDEN_SET_VENDORING.md L30/L32
forbids user uploads to vendors anyway) and becomes the seed/oracle for the SHIPPED BASELINE
('batteries') corpus only. That fits the existing licensed/owned-only constraint perfectly:
Roora labels the founder's collection + ShuttleSet + purpose-recorded clips to build the
baseline, and the 'batteries upgrade' cadence (<5 videos to good quality) is exactly a baseline-
quality metric Roora's labels can certify. Vendor spend stays bounded (one baseline corpus, not
N user corpora).",
      "The minors-exclusion and per-user-training-opt-in guardrails (CLAUDE.md;
OWNED_MODEL_IMPLEMENTATION_PLAN.md M0.3) are EASIER to honor under personalization-first: a
single-user, on-device, opt-in-by-default-to-your-own-data model has a far simpler consent
surface than a pooled training set (no 'split data by match', no cross-user contamination, no
consent ledger across strangers). The per-clip consent ledger (C14) collapses to a per-user
toggle for the personal model."
    ],
    "cons": [
      "The correction loop as a LABEL SOURCE has a built-in selection-bias trap that the
harness's honesty machinery does not fully catch. Users correct what they SEE and care about
(false positives in the reel, a missed favorite rally) -- they do not exhaustively label all
negatives/dead-time. That yields a biased, incomplete label distribution: pseudo-label-then-
correct only surfaces errors near the current model's decision boundary, so the per-user model
can converge to a comfortable local optimum and stop discovering its own systematic blind spots.
LOVO (EXPERIMENT_HARNESS.md 5.1) measures lift HONESTLY but cannot conjure labels for cases the
user never reviewed.",
      "Catastrophic small-N: a hobbyist at 10-15 videos is far below where the project's own
guardrails trust a learned model. MULTI_SIGNAL_FUSION_PLAN.md 3.2 is explicit -- 'until the
labeled set is larger, treat the learned weights as calibration/direction-check, NOT a fully
general model', and OWNED_MODEL_IMPLEMENTATION_PLAN.md warns a bare LOVO mean at n~6 is 'not
promotable on its own'. Per-user F1 deltas at n=10-15 will have wide bootstrap CIs (the honest
?F1 CI + sign test from PR #127 will often straddle 0), so 'turn the person cue OFF for THIS
user' is a decision made on statistically thin evidence -- a real risk of locking in noise as a
per-user default.",
      "Per-user cue decisions can degrade silently and asymmetrically. 'Person cue gives no F1
win for THIS user -> default OFF' is sound when measured, but the person cue's whole job in the
design (MULTI_SIGNAL_FUSION_PLAN.md 1, 4) is to kill the held-shuttle/warm-up false positive
structurally. If a user's early corpus happens not to contain that failure mode, the harness
sees 'no lift' and disables the guard -- then the user records a session full of between-point
shuttle flicks and the model has no defense. Per-user ablation optimizes for the past corpus,
not the next video.",
      "The privacy win is real ONLY on-device, and on-device is explicitly DEFERRED
(OWNED_MODEL_IMPLEMENTATION_PLAN.md L19-20: 'on-device is LATER... do not over-index on it now';
on-device 4B latency on 8GB is an unmeasured open risk). For the near-term hosted-API reality
(COMMERCIALIZATION.md premise: users upload to a cloud GPU VM), 'personalization' means per-user
video + per-user labels SITTING IN THE CLOUD and being trained on -- which RE-OPENS exactly the
DPA / retention / cross-border / Art. 9 surface the pivot claims to escape. The pivot's privacy
thesis writes a check that only the deferred on-device milestone can cash.",
      "Provenance/audit story for per-user weights is messy. The base model's pretraining
provenance is 'unauditable for every open VLM incl. Qwen3-VL' and requires explicit owner risk-
acceptance (OWNED_MODEL_IMPLEMENTATION_PLAN.md L30-32). Personalization MULTIPLIES the artifacts
that inherit that un-auditable base: instead of one model to risk-accept and document, there are
N per-user fine-tunes, each a derivative, each needing the same training-data hygiene (no
minors, opt-in honored) enforced at the edge where it is hardest to audit centrally.",
      "Roora's reframed role removes the only honest cross-user generalization check. Today
Roora-labeled licensed footage is the corpus that proves the model works beyond the founder's
own videos. If Roora only seeds the baseline and every user's quality is measured on their own
correction-loop labels, there is NO independent, vendor-graded held-out set telling you whether
the BASELINE ('batteries') is actually good before personalization kicks in -- you would be
```

certifying the baseline on the same biased per-user signal you are trying to improve. The baseline still needs a Roora-graded generic golden set, so the vendoring spend does NOT actually shrink the way 'just seed the baseline' implies."
    ],
    "findings": [
      "The data substrate for this pivot is REAL and shipped, not aspirational: the correction-loop columns (human_label, confirmed_segment_id) exist in database.py:109-110, the pseudo-label flow (query_candidate_segments(only_unlabeled=True), database.py:299) exists, and the entire per-user experiment loop (Variant + compare + leave_one_video_out + record_experiment, with bootstrap CI + paired sign test on ?F1) shipped 2026-06-13/14 per EXPERIMENT_HARNESS.md and recent git history (commits d7b1f51, #124-127). The owner's claim that 'the harness + eval + A/B were designed to enable this roadmap' is materially supported by the code.",
      "The privacy/licensing guardrails this dimension must honor are DESIGN-ONLY in code today: consent/opt-in/minors-exclusion appear as reserved fields in the M0.1 corpus contract (OWNED_MODEL_IMPLEMENTATION_PLAN.md L103-107: source/consent/training_opt_in/split-by-match) and as CLAUDE.md hard rules, but a grep of database.py finds NO consent/opt-in/minor columns in the live schema -- only the M0.1 plan reserves them. Personalization-first does not relax these guardrails, but it does shift WHERE they must be enforced (per-user, at the edge) and they are not built yet.",
      "The live no-paid-LLM-training VIOLATION (backend/eval/distill_local.py trains on Gemini-silver labels; OWNED_MODEL_IMPLEMENTATION_PLAN.md M0.3(i)) is the same blocker under either roadmap, but personalization-first makes the FIX cleaner: the correction loop gives a first-party human-label source per user, so the per-user training path never needs Gemini-silver at all -- it retargets to human + open-teacher labels by construction. Generalization-first still needs a large pooled silver/golden set and is more tempted by the Gemini shortcut.",
      "Roora's role genuinely shifts from per-user oracle to baseline-corpus oracle, and the existing constraint 'licensed/owned video only, no user uploads to vendors' (GOLDEN_SET_VENDORING.md L30/L32, COMMERCIALIZATION.md C8) makes that the ONLY workable framing -- you cannot legally/privately vendor users' private footage anyway. But the baseline still needs an independent vendor-graded golden set to certify 'batteries' quality, so this does NOT eliminate the generic golden set; it repurposes Roora's output as a baseline-certification + pseudo-label-seeding corpus rather than a per-user ground truth.",
      "Privacy is conditional on the deferred on-device milestone. STORAGE_SHARING_MODEL.md (on-device end-state, L131-135) is the only configuration where 'data never leaves the machine' holds; the current hosted-API premise (COMMERCIALIZATION.md) means near-term personalization trains on cloud-resident per-user video, which re-incurs DPA/retention/Art.9 -- so the data-strategy sequencing must be: ship personalization on-device, or accept that early personalization is NOT the privacy story the pivot advertises."
    ],
    "shapes_future": [
      "Data strategy bifurcates into two corpora with different governance: (1) the BASELINE corpus -- licensed/owned + Roora-graded, the 'batteries', governed centrally with the existing consent ledger / minors-exclusion / split-by-match discipline (M0.1); and (2) PER-USER corpora -- the user's own footage + correction-loop labels, governed by a single per-user opt-in toggle, ideally never leaving the device. These must be kept provably separate (a baseline 'training view' that contains zero user rows, mirroring the M0.1 'no Gemini-sourced rows in any training view' acceptance test).",
      "The correction loop must be upgraded from 'fix what you see' to a deliberate per-user labeling UX that solicits negatives / dead-time / boundary cases, or the per-user label distribution stays biased and the per-user model overfits its own comfort zone. This is a concrete product+data ask: active-learning-style prompts ('was this 3-second clip a rally? yes/no') to balance the label set, feeding the same human_label column. Without it, per-user cue ablation runs on a skewed sample.",
      "Per-user cue-ablation decisions need a safety floor: never disable a STRUCTURAL guard (Gate A net-crossings / held-shuttle defense) on 'no measured lift' alone when the user's corpus lacks that failure mode -- gate ablation on a minimum-evidence bar (the PR #127 ?F1 CI clears 0 AND the sign test agrees AND the failure mode is actually represented in their corpus). Encode 'OFF-for-this-user' as reversible config (EXPERIMENT_HARNESS.md: rollback = flip a flag), and re-evaluate as the user's corpus grows past small-N.",
      "Roora's spend and brief get re-scoped: fund Roora to (a) certify each 'batteries upgrade' baseline against a held-out generic golden set, and (b) produce pseudo-label SEED labels that bootstrap a new user's first <5 videos before their own corrections accumulate. The ROORA_LABELING_GUIDELINES stay as-is (rally [start,end), ending_reason taxonomy) -- they already produce the right artifact; only the consumer changes from 'per-user oracle' to 'baseline oracle + cold-start seeder'.",

```
      "Privacy commitments must be sequenced honestly in the data strategy: do NOT advertise
'data never leaves your machine' until the on-device milestone ships. Near-term cloud
personalization needs the full DPA / retention-limit / minors-exclusion / Art.9 posture as if it
were generalization-first, plus a per-user delete-and-retrain guarantee. The on-device end-state
is what eventually retires that burden -- so on-device moves up the priority list specifically
BECAUSE it is what makes the personalization privacy claim true.",
      "The eval north-star metric should add a per-user dimension: 'videos-to-good-quality' (the
<5 target) becomes a tracked baseline-quality KPI alongside golden-set F1, and the experiment
leaderboard (eval_baselines/experiment_leaderboard.json) should record per-user variant
decisions so 'which cue was turned off for whom, on what evidence' is auditable and reversible."
    ],
    "open_questions": [
      "Where does per-user training physically run in the near term -- on the cloud GPU VM (re-
incurring DPA/retention/Art.9 on per-user video) or only on-device? The privacy thesis of the
pivot is only true in the on-device case, which is explicitly DEFERRED. What is the bridge plan
for hobbyists before on-device exists?",
      "How is per-user label selection bias measured and corrected? The correction loop surfaces
errors near the current boundary; without soliciting negatives/dead-time, what stops the per-
user model from overfitting its own comfort zone and the per-user cue-ablation from running on a
skewed sample?",
      "What is the minimum-evidence bar to set a per-user cue default OFF, given n=10-15 yields
wide ?F1 CIs (the project's own n~6 'not promotable on its own' standard)? And is a structural
guard (Gate A held-shuttle defense) ever allowed to be disabled on 'no lift' when the user's
corpus simply lacks that failure mode?",
      "Does the baseline ('batteries') still get an independent Roora-graded generic golden set
for certification, or does personalization tempt the team to certify the baseline on biased per-
user signal? If the former, the vendoring spend does NOT shrink -- is that budgeted?",
      "How are N per-user fine-tunes governed against the minors-exclusion + opt-in guardrails
and the unauditable-base-model risk-acceptance, when each is a derivative living at the edge
rather than in one centrally-auditable training set?",
      "Does the per-user opt-in default to ON (train on your own data) or OFF? CLAUDE.md
requires a SEPARATE explicit opt-in for per-user training data -- does 'continually self-
optimizes to your footage' as the core product promise conflict with an opt-in-OFF default, and
how is that reconciled for the professional archetype with 100+ hours?"
    ]
  },
  {
    "dimension": "The \"batteries upgrade\" mechanism: how to improve the shipped baseline so a
NEW user reaches superior quality in <5 videos, without a central lake of user video ? plus how
to version the baseline and preserve each user's personalization across an upgrade, and how to
measure the \"<5 videos\" claim.",
    "pros": [
      "Clean conceptual split already exists in code: the shipped 'battery' = a versioned,
hashable artifact (training/gen0/harness.py emits <out>/<version>/{weights.json,manifest.json}
with weights_sha256, git_sha, gt_hashes, feature_schema_version), while per-user tuning = a
Variant recipe (backend/eval/experiment.py: spec_hash, classifier_model path,
min_crossings/min_players/tune_threshold/calibrate). Upgrading the battery = shipping a new
artifact version; personalization = a separate per-user Variant + per-user LogisticRegression
JSON. The two are already decoupled artifacts, so 'upgrade the baseline without wiping a user's
tuning' is mechanically natural: bump the artifact version, keep the user's Variant/model file.
No new platform needed.",
      "The privacy-preserving upgrade sources fall out of guardrails already ratified, in
increasing privacy cost: (1) Roora-vendor-labeled + owner-recorded SEED corpus (the n=6 owned
set, 262 rallies, Proprietary) ? zero user-privacy cost, fully owned, already the harness input;
(2) pooled OPT-IN contributed data ? the schema ALREADY reserves the consent gate
(OWNED_MODEL_IMPLEMENTATION_PLAN M0.1 row field `consent/training_opt_in`, and CLAUDE.md
mandates 'per-user training data needs a separate explicit opt-in; exclude minors'); (3)
FEDERATED updates that ship only per-user LogisticRegression weight vectors (a tiny JSON: numpy
coeffs + frozen normalization via classifier.to_dict), never raw video ? a federated average
over opted-in users' coefficient vectors is a few-KB aggregation, not a video lake; (4)
synthetic augmentation on stored TRAJECTORIES (CSV of x,y,visibility per frame), not pixels ?
perturb/jitter/time-warp trajectories + label windows, which is privacy-trivial because
trajectories are not biometric and never leave as pixels.",
      "Federated averaging is unusually cheap here precisely because the model is a 5-feature L2
LogisticRegression (FEATURE_NAMES = duration, peak_velocity, mean_velocity, active_ratio,
```

```
net_crossings), not a deep net: aggregating per-user coefficient vectors is a weighted mean of
~6-dim vectors ? trivially auditable, no secure-aggregation infra strictly required for a v1
(though it helps), and it composes with the existing train_final()/LOVO flow. The 'aggregate
updates not raw video' privacy property is real and not hand-wavy at this model scale.",
      "The <5-videos metric has an exact, already-built home: a HELD-OUT-USER (really held-out-
MATCH, since split-by-match is enforced) benchmark = a learning curve. For a brand-new user's k
videos, train on k of theirs + eval on the rest, sweep k=1..K, and compare the curve under
battery v_old vs v_new. The harness already does the honest cross-video step
(calibration.leave_one_video_out picks config on others, scores on held-out; harness.sign_test
gives the paired per-video sign test; experiment.honest_summary gives the paired ?F1 bootstrap
CI + sign test). 'Reaches superior quality in <5 videos' becomes a concrete, falsifiable
statement: the k at which the new battery's held-out F1@tIoU0.5 crosses the ship target
(NEXT_STEPS suggests ?0.70 multi-court) drops below 5, with the per-k ?F1 CI clearing 0 and the
sign test agreeing ? exactly the promotion-grade evidence the leaderboard already stores
(honest_delta_f1).",
      "Versioning + personalization preservation is solvable with what's there: pin
(battery_version, feature_schema_version) into each user's Variant record; an upgrade that keeps
FEATURE_SCHEMA v1 lets the user's per-user LogisticRegression and per-user thresholds/cue-flags
carry over byte-identically (the no-regression invariant test pattern: all new-cue flags OFF ?
predictions byte-identical to control). The CONTROL variant being pinned/frozen means rollback
is a config flip, never a git revert ? so a bad battery upgrade can't strand a user. Per-user
cue decisions ('motion cue gives no F1 win for THIS user ? default OFF') survive because they
live in the user's Variant.cue_flags/min_players, independent of the battery weights.",
      "Cadence design has a natural trigger already in the data flow: the user's CORRECTION LOOP
(DB candidate_segments.human_label + confirmed_segment_id) is the per-user label stream, and the
experiment leaderboard is the system-of-record for 'did a change help on which slices.' So a
battery upgrade is gated, not scheduled-blindly: re-run the held-out-user benchmark across the
pooled/seed corpus + synthetic, promote a new artifact version ONLY when run_regression.py (exit
0/1/3) shows no regression on the golden set AND the held-out-user learning curve improves with
CI clearing 0. This is the existing promotion gate, reused ? upgrade cadence = 'whenever a
candidate battery clears the gate,' bounded by a slow human-review cadence (e.g. quarterly) so
users aren't churned."
    ],
    "cons": [
      "The n is brutally small and that is the dominant risk, not the mechanism. The whole
'superior quality' claim today rests on n=6 matches where NOTHING is significant: the Gen-0 sign
test is 5/6 wins at p=0.109 (harness gate_verdict explicitly refuses ship), and the
leaderboard's two recorded fusion experiments are 2/6 (p=0.6875, ?F1 CI [-0.16,+0.16], a LOSS)
and 4/6 (p=0.6875, ?F1 CI [-0.05,+0.20], crosses 0). A held-out-USER learning curve at k=1..4
per user, over a handful of seed users, will have CIs so wide that '<5 videos reaches superior
quality' is unprovable for a long time. The metric is sound; the data to power it does not exist
yet, and no upgrade mechanism manufactures statistical power.",
      "Federated learning on per-user LogisticRegression coefficients leaks more than it looks.
With a 6-dim coefficient vector and a handful of opted-in users, a contributed weight update is
a near-fingerprint of one user's rally distribution; naive FedAvg over few users is a known
privacy/membership-inference surface, and the repo has NO secure-aggregation, NO DP-noise, NO
minimum-cohort threshold today. The privacy 'win' over a video lake is real for the RAW PIXELS,
but selling FL as 'private' without DP + a k-anonymity cohort floor + minors-exclusion
enforcement (the schema RESERVES the consent flag but nothing ENFORCES it in a federated path)
would be overclaiming. This is a counsel/owner-gated item, not an engineering free lunch.",
      "Personalization-first FIGHTS the upgrade goal. The pivot's premise is that wins are tuned
to THIS user and may be weaker on others ? but the battery-upgrade benchmark is a HELD-OUT-USER
metric, i.e. it rewards GENERALIZATION. Improving 'works great in <5 videos for a stranger' is
literally the generalization-first objective wearing a personalization hat. The two seed sources
that most cheaply move the held-out-user curve (owner+Roora seed, pooled opt-in) are generic
data; the per-user correction loop barely feeds the BASELINE (you can't centralize it without
the lake you're avoiding). So the correction loop's value is ~90% local personalization and only
~10% (via FL gradients) baseline upgrade ? the upgrade engine is really the seed/pooled corpus,
and the doc should say so plainly.",
      "Synthetic trajectory augmentation has a low quality ceiling and a measurement trap.
Jittering/time-warping stored WASB CSVs cannot synthesize the FAILURE MODES that actually cost
F1 on the target distribution ? multi-court background-game pickups and dead-time merges
(NEXT_STEPS names these as the exact contrast-metric confound at wrapper 0.66 on Kushagra).
Augmentation that only resamples clean single-court trajectories will inflate LOVO on easy folds
and do nothing for the hard multi-court case that defines the ship target ? and could quietly
```

bias the held-out-user benchmark optimistic if synthetic videos leak into eval folds. Synthetic must be train-only, match-split-clean, and validated to move the HARD folds or it is theater.",
      "Versioning across a FEATURE_SCHEMA change breaks the 'preserve personalization' promise. The clean carry-over only holds while feature_schema_version is stable; the moment a battery upgrade adds/changes a feature (e.g. fusion P3 person features, or M0.4's ActionFormer swap that abandons the 5-feature LogReg entirely), every user's pinned per-user LogisticRegression is dimensionally invalid and their tuning CANNOT be carried byte-identically ? it must be RE-FIT from their stored correction-loop labels. That re-fit needs the user's labeled candidates persisted and re-runnable per user, and a migration path; none of that is designed yet. The doc must distinguish a 'minor' battery upgrade (schema-stable, weights-only, tuning preserved) from a 'major' one (schema change, per-user re-fit required).",
      "There is no per-user model store, no per-user benchmark runner, and no upgrade orchestration today ? only the single-corpus harness + a single global leaderboard + a single regression_baseline.json. The architecture is explicitly single-tenant/local-first (EXPERIMENT_HARNESS §9: 'NOT a service'). Federated aggregation, an opt-in consent ledger, a per-user artifact registry, and a held-out-USER (multi-user) benchmark are all genuinely NEW surfaces that cut against the stated 'keep it the size of baseline.json' guardrail. The mechanism is designable on these primitives, but it is NOT '80% already built' the way the single-user experiment harness was."
    ],
    "findings": [
      "The codebase already cleanly separates the two artifacts an upgrade must touch independently: the BATTERY = a versioned bundle from training/gen0/harness.py (`<out>/<version>/weights.json` + `manifest.json`, carrying weights_sha256, git_sha, feature_schema_version, per-video gt_hashes, and the LOVO eval + ship-gate verdict), and the PERSONALIZATION = a Variant (backend/eval/experiment.py: spec_hash + a per-user classifier_model JSON path + min_crossings/min_players/cue_flags/tune_threshold/calibrate). Upgrade-without-wipe is therefore an artifact-versioning problem, not a retraining-from-scratch problem ? as long as feature_schema_version is held stable.",
      "The privacy-preserving upgrade sources map exactly onto existing guardrails and data shapes, ordered by privacy cost: (1) owner+Roora SEED corpus (already the harness input, Proprietary, n=6/262 rallies ? zero user-privacy cost); (2) pooled OPT-IN data, gated by the schema field M0.1 already reserves: `consent/training_opt_in` + `source(human|pseudo|vendor)`, with CLAUDE.md's 'separate explicit opt-in; exclude minors' rule; (3) FEDERATED averaging of per-user LogisticRegression COEFFICIENT vectors (a ~6-dim numpy JSON via classifier.to_dict ? a few KB, never pixels); (4) synthetic augmentation on stored TRAJECTORY CSVs (x,y,visibility per frame), not video. Only (1)+(4) are privacy-trivial today; (2)+(3) are owner/counsel-gated.",
      "The <5-videos metric is a HELD-OUT-USER (held-out-MATCH, since split-by-match is enforced) LEARNING CURVE, and the harness already computes every piece: for a new user's K videos, sweep k=1..K, train on k, eval on the rest with calibration.leave_one_video_out (honest cross-video pick), and compare the curve under battery v_old vs v_new using the SAME promotion-grade stats the leaderboard already stores ? experiment.honest_summary's paired ?F1 bootstrap CI + exact sign test (eval_stats) and harness.sign_test. 'Superior in <5 videos' = the smallest k where new-battery held-out F1@tIoU0.5 crosses the ship target (NEXT_STEPS proposes ?0.70 multi-court) with the per-k ?F1 CI clearing 0 AND the sign test agreeing.",
      "DOMINANT BLOCKER is statistical power, not design: at the current n=6, nothing is significant. The Gen-0 harness itself refuses to ship (sign test 5/6, p=0.109) and the two recorded fusion experiments in eval_baselines/experiment_leaderboard.json are a 2/6 LOSS (p=0.6875) and a non-significant 4/6 (p=0.6875, ?F1 CI [-0.054,+0.203] crosses 0). A held-out-USER curve over a few seed users at k=1..4 will have CIs too wide to prove '<5 videos' for months. The honest framing: the upgrade MECHANISM is buildable on existing primitives, but the CLAIM is currently unmeasurable ? corpus growth is the gating input, exactly as NEXT_STEPS step 1 already says.",
      "Two upgrade CLASSES must be distinguished or the 'preserve personalization' promise breaks: a MINOR upgrade keeps FEATURE_SCHEMA v1 ? ship new battery weights, carry each user's per-user LogisticRegression + thresholds + cue_flags byte-identically (the existing no-regression invariant pattern: flags OFF ? byte-identical to control), rollback = config flip on the pinned CONTROL variant. A MAJOR upgrade changes the feature schema (fusion P3 person features, or the planned M0.4 ActionFormer/ASFormer swap that drops the 5-feature LogReg) ? every per-user model is dimensionally invalid and MUST be re-fit from that user's stored correction-loop labels (candidate_segments.human_label / confirmed_segment_id), requiring a per-user re-fit migration that does NOT exist yet.",
      "Federated learning's privacy claim needs hardening before it can be sold as 'private': with a ~6-dim coefficient vector and few opted-in users, a contributed update is close to a per-user fingerprint (membership-inference surface), and the repo has no DP noise, no secure

aggregation, and no minimum-cohort (k-anonymity) floor, and the consent/minors flag is RESERVED but not ENFORCED on any federated path. The raw-pixel privacy win is genuine; the gradient-privacy win is not free and is correctly an owner/counsel-gated item, consistent with the project's licensing/consent posture.",

"The CORRECTION LOOP feeds personalization ~90% and the baseline only ~10%: it is the per-user label source (DB human_label / confirmed_segment_id) that drives each user's own Variant/threshold/cue selection, but it cannot move the SHARED baseline without either the central video lake the pivot rejects, or the federated-gradient path (which only contributes a small, noisy coefficient nudge). So the real baseline-UPGRADE engine is the SEED + POOLED-OPT-IN + SYNTHETIC corpus; the held-out-user benchmark is, ironically, a GENERALIZATION metric ? improving it is the generalization-first objective. The doc should state this honestly rather than implying the correction loop upgrades the battery."

],

"shapes_future": [

"Decide the upgrade taxonomy FIRST: ratify a MINOR (feature-schema-stable, weights-only, per-user tuning carried byte-identically, rollback = config flip) vs MAJOR (schema change ? per-user re-fit required) distinction, because the planned M0.4 ActionFormer/ASFormer swap is a MAJOR upgrade that invalidates every per-user 5-feature LogisticRegression. Until the per-user re-fit migration (re-train from stored candidate_segments.human_label/confirmed_segment_id) exists, no schema-changing battery upgrade can preserve personalization ? this constrains the M0.4 roadmap.",

"Build the held-out-USER benchmark as the gate for ANY battery release: a learning-curve runner that, for each seed/opted-in user, sweeps k=1..K via calibration.leave_one_video_out and reports the per-k held-out F1 + paired ?F1 CI + sign test (reusing experiment.honest_summary / harness.sign_test) for v_old vs v_new. Promote a new battery version ONLY when run_regression.py stays exit 0 on the golden set AND the curve's crossing-k drops with CI clearing 0. This makes '<5 videos to superior quality' a falsifiable release criterion, not a slogan.",

"Sequence the four upgrade sources by privacy cost and build them in that order: (1) seed/owner+Roora corpus growth (do now ? it is NEXT_STEPS step 1 and gates everything), (2) train-only synthetic trajectory augmentation validated to move the HARD multi-court folds (must be match-split-clean, never in eval), (3) pooled opt-in (needs the consent ledger that ENFORCES the reserved consent/training_opt_in + minors-exclusion flag), (4) federated coefficient averaging (needs DP noise + a minimum-cohort floor + counsel sign-off). Treat (3)+(4) as owner/counsel-gated, consistent with the project's licensing/consent discipline.",

"Make the per-user artifact a first-class, versioned record: store per user the tuple (battery_version, feature_schema_version, per-user classifier JSON, thresholds, cue_flags) and pin the battery_version + spec_hash, so an upgrade is auditable and a bad battery is rollback-by-config on the frozen CONTROL variant. This extends the existing single global leaderboard + regression_baseline.json pattern to a per-user registry WITHOUT becoming MLflow/W&B ? staying within EXPERIMENT_HARNESS §9's 'size of baseline.json' guardrail.",

"Re-frame the correction loop's role honestly in the pivot doc: it is the PERSONALIZATION label source (drives each user's own cue selection / threshold / per-user model), and only a SECONDARY, federated-gradient contributor to the shared battery. The battery-upgrade engine is the seed + pooled + synthetic corpus, and the held-out-user benchmark it optimizes is a GENERALIZATION metric ? so the 'batteries upgrade' and 'generalization-first' goals are the same objective viewed from two angles; the personalization win lives entirely in the per-user Variant layer on top.",

"Gate cadence on evidence, not the calendar: a candidate battery may be PREPARED continuously (as corpus/synthetic/opt-in data lands) but RELEASED to users only when it clears the held-out-user gate, then rolled out behind the pinned-control rollback so a regression is a config flip. Bound user-facing churn with a slow human-review release cadence (e.g. quarterly), and require that each released battery's manifest carry the held-out-user curve + the n it was measured at, so the '<5 videos' claim always ships with its honest CI."

],

"open_questions": [

"At what corpus size N (matches) does the held-out-USER learning curve at k<5 produce a ?F1 CI that clears 0 with a significant sign test? Today n=6 gives p?0.11-0.69 on every recorded comparison ? what is the minimum N (and number of distinct seed users) before '<5 videos to superior quality' is even measurable, let alone provable?",

"Does the planned M0.4 ActionFormer/ASFormer swap (which abandons the 5-feature LogisticRegression) count as a MAJOR battery upgrade requiring a per-user re-fit, and is the project willing to build the per-user re-fit-from-correction-labels migration before that swap, or does personalization get reset on that one upgrade?",

"For the federated path: what minimum opted-in cohort size, DP-noise budget, and secure-aggregation requirement does the owner/counsel require before per-user coefficient updates can

```
be aggregated ? and who enforces the reserved consent/training_opt_in + minors-exclusion flags
on that path (today they are schema fields, not enforced gates)?",
      "Is synthetic trajectory augmentation actually able to move the HARD multi-court folds
(background-game pickups + dead-time merges, the wrapper-0.66 confound), or only the easy
single-court folds? Without that validation, synthetic data risks optimistically biasing the
very held-out-user benchmark used to justify the upgrade.",
      "Should the held-out unit be the USER or the MATCH? Split-by-match is enforced today, and
a single user contributes multiple matches ? does 'held-out user' mean hold out ALL of one
user's matches (true new-user simulation, fewer effective folds) or hold out matches across
users (more folds, weaker new-user claim)? The two give different, both-defensible numbers for
the '<5 videos' headline.",
      "How is a per-user personalization regression detected at upgrade time? run_regression.py
guards the shared GOLDEN set, but a battery upgrade could improve aggregate held-out-user F1
while DEGRADING specific power-users (archetype-b, 100+ hours) whose tuning was finely matched
to the old battery ? is there a per-user no-regression check, and what is the rollback unit
(global battery vs per-user opt-out)?"
    ]
  },
  {
    "dimension": "Deep-analysis / shot-taxonomy feasibility (serves, smashes, drives, drop
shots, clears) under the personalization-first pivot",
    "pros": [
      "Clean phase separation already exists to layer on: HOW_RALLY_DETECTION_WORKS.md keeps
DETECTION (where is the shuttle/who is on court) separate from DECISION (windowing), and shot-
type classification is naturally a *third* layer that consumes the same persisted per-frame
trajectory + person tracks. Adding it does not require touching the frozen WASB model.",
      "The persist-once / re-score-forever substrate (candidate_segments.stats JSON,
EXPERIMENT_HARNESS.md §2) generalizes cleanly from rally labels to shot labels: the same offline
LOVO machinery (compare(), leave_one_video_out, the logistic-regression artifact, the no-
regression gate) can grade a per-shot variant with NO new scoring math ? the harness was
genuinely designed for additive cues.",
      "Per-user shot-type tuning is *more* plausible than per-user rally tuning for the pro
archetype (b): a professional with 100+ hours and a correction loop generates many labeled
smashes/drops/clears, so within-user class balance and N stop being the bottleneck ? the
personalization thesis (deep analysis of THIS player's shot mix) is strongest exactly where data
volume is highest.",
      "Some shot structure is already cheaply derivable from signals the pipeline computes: peak
shuttle speed (already a rally feature), shuttle arc/height, and direction-reversals (each
reversal ? a racket contact, noted in HOW_RALLY_DETECTION_WORKS.md §'sparks') give a coarse
smash-vs-drop / shot-cadence signal WITHOUT any new model ? a low-cost first rung before full
taxonomy.",
      "shots_count is already a golden-label column (ROORA v8 §1) and Appendix B Tier-B already
scopes per-shot timestamps + per-shot type {serve, clear, drop, smash, net, drive, lift} as a
priced vendor extension ? so the labeling pipeline to BOOTSTRAP a shipped baseline ('batteries')
exists on paper; per-user correction then refines it.",
      "Shot taxonomy is the natural unit for the product story (deep analysis of a user's
serves/smashes/drives) and for the standing paper aim (QUALITY_ITERATIONS.md) ? a per-shot
ablation backbone is a strong differentiator vs commodity rally clipping."
    ],
    "cons": [
      "Cold-start is categorically harder than for rallies. Rally detection is a 1-class
temporal problem with ~10-15 rallies per short video; shot taxonomy is a 6-7-class problem where
each rally yields a handful of shots but they are dominated by serves + clears + drives, so
smashes/drop-shots/nets are the MINORITY classes. A hobbyist's 10-15 videos can plausibly
contain very few labeled smashes ? severe class imbalance, not just small N.",
      "The repo has ZERO shot-taxonomy capability today ? grep across the whole codebase finds
shots_count only as a CSV header in one test fixture and NO direction_reversal / shot-type code
at all. The 'batteries-included baseline that needs <5 videos' would be built from scratch;
there is no shipped shot model to self-optimize, unlike rallies where a heuristic + Gen-0 head
already exist.",
      "The correction loop as a per-user shot-label source is weak: today's correction surface
is rally [start,end] confirm/reject (human_label / confirmed_segment in candidate_segments). A
user confirming 'this is a rally' does NOT produce a per-shot-type label. Capturing
serve/smash/drop labels needs a NEW, finer correction UI and asks far more user effort per video
? exactly the friction the hobbyist archetype (a) will not absorb.",
```

"The empirical reality check is sobering: even RALLY detection is not 'heavily optimized by 10-15 videos.' QUALITY_ITERATIONS.md shows shuttle-only LOVO F1 0.307, learned-with-threshold 0.423 at n=6 GOLDEN videos, person cue gives NEGATIVE lift, audio is +0.079 but the CI spans 0 and the sign test is n.s. The standing lesson is explicit: 'Calibration, not representation, is the small-N wall ? more distinct matches is the highest-ROI unblock.' Shot taxonomy multiplies the data hunger that rally detection has not yet satisfied.",
"ROORA v8 itself flags shot-type as needing 'its own short guideline + an inter-annotator agreement check' (Appendix B) ? forced-vs-unforced is already 'the one judgment call,' and smash-vs-clear-vs-drive boundaries are even more subjective. Per-user self-labels via a correction loop will be NOISIER than vendor golden labels, so the per-user training signal is lower quality precisely where data is scarcest.",
"Domain shift hits harder: the static-tripod side/overhead camera (vs broadcast) makes shuttle arc and player pose ambiguous; distinguishing a fast drive from a flat smash or a tight net-shot from a drop requires height/trajectory + possibly pose (MediaPipe/RTMPose, parked as cue #2/#3 in MULTI_SIGNAL_FUSION_PLAN.md §6) ? a second model and more cost, not a free re-score.",
"'Optimized for THIS user, weaker on others' is acceptable framing for a calibration knob, but for shot taxonomy it risks a per-user model that overfits one player's style/court and cannot be honestly evaluated ? LOVO needs multiple of THAT user's matches, and the hobbyist may never have enough held-out matches to prove a per-user shot lift is real rather than noise."
],
"findings": [
"Shot taxonomy is entirely GREENFIELD in this repo: a whole-codebase grep finds no shot-type or direction-reversal detection code; shots_count exists only as a golden-label CSV column (ROORA v8 §1) and as a Tier-B *priced vendor extension* (Appendix B), never as a detected signal. There is no 'battery' to self-optimize yet ? it must be built before personalization can refine it.",
"Per-user shot labels do not fall out of the existing correction loop. Today's human-in-the-loop surface is rally-level (human_label / confirmed_segment_id in candidate_segments, database.py); confirming a rally yields no serve/smash/drop label. A new, finer-grained and higher-effort correction UI is a prerequisite ? a hard ask for the casual hobbyist archetype (a).",
"Class imbalance, not just small N, is the cold-start killer for shot types. Rally detection is effectively single-class with ~10-15 positives per short video; shot taxonomy is 6-7 classes dominated by serves/clears/drives, so the product-relevant minority classes (smash, drop, net) may be near-absent in a hobbyist's 10-15 videos ? 'heavily optimized by 10-15 videos' is realistic for rally/highlight but NOT for full shot taxonomy.",
"The measured rally numbers cap expectations honestly. QUALITY_ITERATIONS.md (n=6 golden): shuttle-only F1 0.307, learned+threshold 0.423; person cue NEGATIVE lift; audio +0.079 but CI [-0.054,+0.203] spans 0, sign test 4W/2L n.s. The explicit standing lesson is that more distinct matches ? not cleverer features ? is the highest-ROI unblock. Shot taxonomy strictly increases this data appetite.",
"A cheap, no-new-model first rung exists: peak shuttle speed (already computed), shuttle arc/height, and direction-reversals (? racket contacts) can give a coarse smash-vs-soft / shot-cadence cut that rides the SAME harness (compare/LOVO/logistic) without pose or a new detector ? this is the realistic entry point before any full taxonomy.",
"Realistic phasing (recommended): Phase 0 (now) ship rally + highlight + the rally-level correction loop and accumulate per-user labels; Phase 1 add a coarse 2-3-bucket shot proxy (hard/soft, derived from speed/arc/reversals) measured on golden via the existing harness, default-OFF; Phase 2 add a finer-grained per-shot correction UI + Tier-B vendor shot labels to build a SHIPPED multi-class 'battery' baseline (generalization-first for the baseline); Phase 3 ONLY for high-volume users (pro archetype b) enable per-user shot-type calibration once enough of THAT user's held-out matches exist to LOVO-prove a per-user lift. The hobbyist gets the shipped baseline, not a self-tuned shot model."
],
"shapes_future": [
"If full per-user shot taxonomy is committed near-term, it forces a NEW per-shot correction UI + a new multi-class label schema (extend candidate_segments / a shots table) and a Tier-B Roora engagement ? a meaningful product + vendor + data-pipeline investment, not a re-score. The pivot decision should not assume the harness alone delivers it.",
"The personalization-first thesis splits by archetype for shot taxonomy: it is defensible ONLY for the high-volume professional (b) where within-user N + class balance can support a per-user shot model; for the hobbyist (a) the honest design is a generalization-first shipped baseline ('battery') with at most rally-level personalization. The roadmap should make this archetype split explicit rather than promising per-user shot tuning to everyone.",

      "Whether 'batteries upgrades ? <5 videos to superior quality' is achievable depends entirely on building a strong GENERALIZED shot baseline first (cross-user vendor labels), which is the OPPOSITE of personalization-first for this layer ? i.e. the deep-analysis layer pulls strategy back toward generalization-first for the baseline, with personalization as a thin per-user calibration on top.",
      "Sequencing matters: investing in shot taxonomy before the rally layer clears its own small-N wall (the explicit 'grow the corpus' top priority in NEXT_STEPS.md) risks stacking an unproven multi-class layer on a still-borderline rally substrate ? phasing rally/highlight first is not just product-nice, it de-risks the data dependency the shot layer inherits."
    ],
    "open_questions": [
      "What is the minimum viable shot taxonomy for v1 ? a coarse 2-3 bucket (hard smash/drive vs soft drop/net vs serve) that the existing speed/arc/reversal signals can support without pose, or does the product story require the full 6-7 class {serve, clear, drop, smash, net, drive, lift}? The feasibility verdict flips sharply between these.",
      "How will per-user shot labels actually be captured ? will users tolerate a per-shot correction UI, and at what effort budget? Without a concrete answer, the 'correction loop as per-user shot-label source' premise is unvalidated for the hobbyist archetype.",
      "For a hobbyist with 10-15 videos, what is the realistic count of labeled SMASHES / DROP-SHOTS (the minority, product-interesting classes)? A quick audit of shots_count + ending_reason distributions in the existing 6 golden matches (262 rallies) would empirically bound whether minority-class N is ever sufficient per user.",
      "Does shot discrimination require pose (MediaPipe/RTMPose, parked cue #2/#3) or can box + trajectory features carry it? This determines whether the deep-analysis layer is a cheap re-score or a second-model cost ? a small offline probe on the golden trajectories (smash vs drop separability from speed/arc/reversals) would answer it before any commitment.",
      "What inter-annotator agreement does shot-type labeling actually achieve (ROORA v8 flags it as needing its own IAA check)? Low agreement on smash-vs-drive-vs-clear would cap the achievable F1 of ANY per-user shot model and should gate whether the layer is worth building.",
      "Can a per-user shot model ever be HONESTLY evaluated for a low-volume user, given LOVO needs multiple held-out matches FROM THAT USER? If not, per-user shot 'optimization' for hobbyists is unfalsifiable and should be dropped from the pivot's claims for that archetype."
    ]
  }
]

Make the STRONGEST possible counter-case: why PERSONALIZATION-FIRST might be the WRONG pivot. Cover at least: (a) TAM/economics ? does optimizing-for-one-user shrink the market vs a generic model with network effects; (b) the "anecdotal wins on other videos" hand-wave ? is per-user overfitting actually a feature or a trap; (c) eval noise ? per-user corpora are even smaller than n=6, so per-user A/B promotions could be noise-driven; (d) compute/maintenance ? on-device training + per-user model stores + per-user upgrades = real burden; (e) cold-start for shot types; (f) is the "harness was designed for this" a genuine advantage or post-hoc rationalization. Then state the conditions under which generalization-first wins instead. Return a frank assessment (free text).

2. assistant (2026-06-14T08:27:41.404Z)

This is a strategy analysis task, not a coding task. The owner wants the strongest possible counter-case against the "personalization-first" pivot. I have rich parallel research already provided. Let me check whether there are project docs that would sharpen the argument with specific numbers, then write a frank assessment. But first, let me verify a few load-bearing empirical claims against the actual repo so the counter-case is grounded in real numbers, not the research summary's paraphrase.

3. user (2026-06-14T08:27:44.984Z)

Already up to date.
---LEADERBOARD---
[
  {
    "timestamp": "2026-06-14T06:23:51.507393+00:00",
    "iou": 0.5,
    "label_set_hash": "50d98ffbc370",
    "num_videos": 6,

```
    "variant_a": {
      "name": "shuttle_only",
      "spec_hash": "47c867c36f2f",
      "aggregate": {
        "precision": 0.31807603780927995,
        "recall": 0.3101947096042718,
        "f1": 0.3068611057861412,
        "mean_iou": 0.5791831889148886
      }
    },
    "variant_b": {
      "name": "shuttle_person_fusion",
      "spec_hash": "253f4709b4c4",
      "aggregate": {
        "precision": 0.3617228001510626,
        "recall": 0.2708939086387243,
        "f1": 0.2863398846047612,
        "mean_iou": 0.5690111885714347
      }
    },
    "delta": {
      "precision": 0.04364676234178266,
      "recall": -0.0393008009655475,
      "f1": -0.02052122118138,
      "mean_iou": -0.01017200034345389
    },
    "honest_delta_f1": {
      "mean_delta": -0.02052122118137999,
      "ci_low": -0.16462806145542527,
      "ci_high": 0.16224446866060574,
      "n": 6.0,
      "ci_excludes_zero": false,
      "sign_test": {
        "wins": 2.0,
        "losses": 4.0,
        "ties": 0.0,
        "n_effective": 6.0,
        "p_value": 0.6875
      }
    }
  },
  {
    "timestamp": "2026-06-14T06:24:56.487711+00:00",
    "iou": 0.5,
    "label_set_hash": "50d98ffbc370",
    "num_videos": 6,
    "variant_a": {
      "name": "shuttle_person",
      "spec_hash": "ee89f1e41604",
      "aggregate": {
        "precision": 0.3617228001510626,
        "recall": 0.2708939086387243,
        "f1": 0.2863398846047612,
        "mean_iou": 0.5690111885714347
      }
    },
    "variant_b": {
      "name": "shuttle_person_audio",
      "spec_hash": "d68c39eba87c",
      "aggregate": {
        "precision": 0.4530798177857001,
        "recall": 0.3408138120839734,
        "f1": 0.3655826552797436,
        "mean_iou": 0.7074802852140344
      }
```

```
    },
    "delta": {
      "precision": 0.0913570176346375,
      "recall": 0.06991990344524907,
      "f1": 0.07924277067498242,
      "mean_iou": 0.13846909664259965
    },
    "honest_delta_f1": {
      "mean_delta": 0.07924277067498244,
      "ci_low": -0.053508315003560535,
      "ci_high": 0.2029045239922802,
      "n": 6.0,
      "ci_excludes_zero": false,
      "sign_test": {
        "wins": 4.0,
        "losses": 2.0,
        "ties": 0.0,
        "n_effective": 6.0,
        "p_value": 0.6875
      }
    }
  }
]
```


4. user (2026-06-14T08:27:45.082Z)

docs\QUALITY_ITERATIONS.md

5. user (2026-06-14T08:27:45.082Z)

eval_baselines\heuristic_lovo_n6.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\experiment_leaderboard.json

6. assistant (2026-06-14T08:27:47.968Z)

Let me read the empirical backbone doc and the cold-start / proposal-recall ceiling claims, which are load-bearing for the counter-case.

7. user (2026-06-14T08:27:48.789Z)

```
1       # Rally-detection quality iterations ? living log
2
3       **Purpose.** A running, reverse-chronological log of how we evolve rally-detection
quality: each
4       iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.
This is the
5       **live** continuation of the historical trail in
6       [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the
archive policy
7       (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
file stays at
8       `docs/` top level and keeps growing, then its entries graduate to the archive when they
conclude.
9
10      **Methodology (non-negotiable).** Every quality change rides the experiment harness
11      ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default
OFF**; they're
12      graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
13      on measured lift** through the `run_regression.py` gate; **rollback = flip a config
flag, never
14      `git revert`.** Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
```

in EVALUATION.md; the 2-layer model in HOW_RALLY_DETECTION_WORKS.md.

Golden corpus = the 6 labelled matches in `output/human_lovo_manifest.json` (owner-confirmed 2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles`). The model trained on these is what we run alongside the existing detector on NEW footage for the side-by-side review.

Baselines that drive strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	0.93	0.66

Reading: clean footage is commoditized (serve via Gemini); **multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target.** Baselines tracked in `eval_baselines/`. The owner's observed false positive ? a player **holding/flicking** the shuttle between points counted as a rally ? is the precision bug the fusion gates target.

Current track (live) ? newest first

Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising (not promotable) *(2026-06-14)*

The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ? `output/*_golden_features.json`, the replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats (C1 ? bootstrap 95% CI + paired sign test; never a bare mean at n=6)**:

Variant (honest LOVO, IoU ? 0.5)	F1	?F1 vs prior	95% CI	sign test	CI clears 0
shuttle-only	0.307	?	?	?	?
+ person (P3)	0.286	**?0.021** (?6.7%)	[?0.165, +0.162]	2W/4L (p=0.688)	**No**
+ person + audio (P4 incr.)	0.366	**+0.079** (+27.7%)	[?0.054, +0.203]	4W/2L (p=0.688)	**No**

Read honestly: neither cue clears the promotion bar (both CIs span 0, both sign tests n.s.) at n=6 ? **both stay default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ? `players_in_court` counts spectators); per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising lead ? +0.079 / +27.7%, wins 4/6 (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above** shuttle-only (0.307) ? but n=6 can't confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and measure audio on shuttle-only *directly* (without the person handicap). Recorded in `eval_baselines/experiment_leaderboard.json`. **Negative/uncertain results are kept on purpose ? this is the honest ablation backbone for the paper aim.**

Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration *(2026-06-14)*

Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness). The measured win (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg 1.68?1.01 ? the [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears the honest bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the eval-only Gate A folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the recall-oriented *proposal* candidate set; flipping the **live served default** stays owner-gated until a real fusion-path run confirms it (the production fusion segmenter is P2/unverified on real video) ? so this records the promotion + rationale, not a silent default flip. **Corroborated on real footage:** offline re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ? CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs (no golden labels for this footage ? divergence, not F1).

Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-indexer/pull/130). The first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff artifact. Added additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train** fold) + `calibrate` (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423 (+0.116)**, testlarge 0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)**. Threshold-in-LOVO is the big lever; calibration is available too.

Next: a real fusion-path run to confirm Gate-A served-side; per-video output layout (PER_VIDEO_OUTPUT_LAYOUT.md) so each video's artifacts are self-contained.

First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB trajectories, no GPU). **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p = 0.031**, CI [+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01**. The window-level 5-feature logistic **did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the stronger per-frame prototype). The framework held up: **C1 promoted one variant and refused another; C4 surfaced over-segmentation tIoU hid.** Full record + tables ? [`archives/quality-iterations/2026-06-first-ab-experiment/`](archives/quality-iterations/2026-06-first-ab-experiment/README.md). Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is the C2 (calibration) + threshold-in-loop task list.

Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF *(2026-06-14)*
Why. A multi-agent literature review (folded into ANALYZERS_AND_RECONCILERS.md §13) confirmed the framework's *skeleton* is standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion ? cite, don't claim),

and flagged that our **measurement** is walking into known small-N traps that would make "promote only on measured lift" promote noise or hide a real win. Four corrections, shipped as **pure, additive helpers** (no existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):
- **C1 ? never a single mean.** `backend/eval/eval_stats.py`: `mean_with_ci`/`bootstrap_ci` (deterministic, seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?" test the corpus already needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean is not trustworthy.
- **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt + isotonic (PAVA) calibrators + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an integer net-crossing count are comparable before the arbiter and the over-confident cue can't swamp the rest.
- **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-one-group-out by match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no leakage).
- **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`: `f1_at_overlaps` (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU), `segment_count_ratio`. tIoU alone is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler exists to fix.

Status. Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free lanes); ruff + mypy clean. **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to report CIs + F1@k is the follow-on, owner-gated). The discipline they encode is the methodology for every future cue measurement ? see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.

Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
Hypothesis. Adjudicating shuttle-seeded windows with the net-crossing **Gate A** (`min_crossings?2`) and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up / one-sided-feed / spectator false positives ? *without* hurting recall, because the shuttle stays the primary, camera-invariant seed.

Scope ? and an honesty boundary from the archive (read this). The [`2026-06-multivideo-lovo`](archives/quality-iterations/2026-06-multivideo-lovo/README.md) iteration already smoke-tested **court-region detection as a fix for the *multi-court contrast* confound** and found it **low-headroom**: only 12?14% of dead-time shuttle detections fall *outside* the foreground court, so a perfect court gate lifts contrast to only ~1.6?2.2× (below the ?3.6× "works" threshold) ? the multi-court problem is **temporal**, and the **learned model**, not a court mask, is its answer. So Gate B here is **NOT** sold as the multi-court fix. It targets a **different, previously-untested** failure mode (held-shuttle/warm-up/one-sided/spectator FPs). Two distinct levers; don't conflate them.

Data support for Gate A. The same archive shows real rallies on the multi-court matches contain **?2 net-crossings 97% of the time (median 5)** while a single service feed crosses ~once ? `min_crossings?2` is a principled, data-backed discriminator (it's the held-shuttle owner-bug killer).

Shipping shape. A NEW `fusion` segmenter, **default OFF** (`default_segmenter` unchanged); Gate A wired into the live trajectory path behind `indexing.min_crossings` (default 0 = OFF ? wasb/tracknet byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B is player-count-only; supply ? masked). Person features (`players_in_court, two_sided_motion, mean_player_motion, in_court_ratio, shuttle_player_colocation`) persisted into candidate `stats` for P3's learned fusion. Reuses `PersonDetector` (P1) + `rally_gate` (already existed) ? no reinvention.

Status (2026-06-14): SHIPPED, default-OFF. `FusionSegmenter` (`backend/pipeline/segmenters/fusion_hybrid.py`, registry key `fusion_hybrid`) + the pure feature math (`backend/pipeline/detectors/fusion_features.py`) + two identity hooks on the trajectory engine (`_enrich_candidate_stats`, `_select_for_handoff`) + `PersonDetector. detections_per_frame` + the `FusionConfig` block. Built design-first (a `fusion-p2-design` fan-out workflow), then **adversarially reviewed by a 31-agent workflow (25 findings ? 19 confirmed)**: the headline confirmation is **NO-REGRESSION verified** ? wasb/tracknet twins persist byte-identical stats and hand off all windows (pinned by the extended equivalence test); default `FusionSegmenter` == its wasb substrate. The confirmed low/medium findings were fixed (frame-dim guards, `round()` frame indices, the per-window axis-estimate redundancy ? a reusable `rally_gate` estimate, honest torch-import docstrings, and 6 new tests closing the person-detector / Gate-B / substrate / colocation-halo gaps). Suite 441?**467 green**; ruff + mypy clean. **Two findings deferred with intent** (reviewer-agreed): person features are computed eagerly even for Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics); and `fusion.velocity_threshold` must be set equal to the substrate's to A/B against a tuned wasb run (documented in `FusionConfig`). **Lift still to be measured** offline on the 6 golden videos via `experiment.compare(CONTROL, fusion)`; the held-shuttle-drop + person-feature values are GPU/owner-side. *(Measured numbers appended here when run ? no claims before.)*

Person cue extracted to a reusable producer (fusion P1) ? [#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13

Lifted the torchvision detector + ByteTrack + velocity windowing out of `yolo_hybrid` into `backend/pipeline/detectors/person_detector.py::PersonDetector`. **No behavior change** (proven by the unchanged `process_video` output tests); `yolo_hybrid` now orchestrates only. **Why it's a quality move:** it makes "add the person cue to fusion" an *add-a-producer*, not a segmenter rewrite. Suite 440?441 green.

Experiment harness: A/B without rollback risk ? [#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13

Shipped `backend/eval/experiment.py` ? the thin variant/experiment layer (the audit's "~80% already exists" held; pure glue over `metrics`/`calibration`/`classifier`/`training`/`regression.gt_hash`). Gives `Variant` (+ pinned `CONTROL`), `compare()` (labeled A/B, LOVO-honest), **`compare_unlabeled()`** (the SxS read on NEW unlabeled footage ? agreed / A-only / B-only), and a JSON experiment

leaderboard. **This is
173   the instrument every iteration is measured on.** 19 tests incl. the no-regression invariant (all cues OFF
174   ? byte-identical to control). Suite 421?440 green.
175
176   ---
177
178   ## Archived predecessors (the historical trail)
179   Completed iterations live under [`archives/quality-iterations/`](archives/quality-iterations/README.md):
180   - **`2026-06-initial/`** ? seed of the quality phase.
181   - **`2026-06-human-gt-tuning/`** ? first eval vs **human** GT on a partial-label video; partial-label
182   harness + labeled-window methodology; baseline 0.650 ? tuned 0.742 (*fit-on-self*); found+fixed a
183   negative-start ingestion bug (collector PR #13).
184   - **`2026-06-multivideo-lovo/`** ? the **n=6 corpus + honest LOVO**; established the heuristic is
185   **footage-fragile** (multi-court Kushagra 0.16 / gBaaddy 0.28), the **"contrast" metric** predicts
186   heuristic F1 monotonically, and a **learned per-frame prototype overtook the heuristic at n=3 and holds
187   ~+0.10** (n=6: learned 0.583 vs heuristic 0.480), with **AUC 0.83?0.92 decoupled from contrast**. This
188   is the data-backed green light for the owned model.
189
190   ---
191
192   ## Standing lessons (carry-forward)
193   - **Persist detection once, re-score variants forever.** Expensive/risky = per-frame detection (GPU/WSL);
194   cheap/swappable = the decision. Most experiments never enter the served pipeline.
195   - **LOVO or it didn't happen.** Score on a held-out *video*, never a held-out window from the same video.
196   Group by **match**, not file. Sign test needs n ? ~n=10/9-wins for p<0.05.
197   - **The multi-court confound is TEMPORAL, not spatial/loudness** ? three heuristics were tested and failed
198   for the same reason (foreground spatial gate **?0.04**; audio energy **AUC ~0.6**; court-region
199   separability **12?14% out-of-region**): the dead-time confusion is rally-vs-non-rally activity in the
200   *same place and frequency band*. Only the learned model's temporal/structural features crack it (AUC
201   0.83 on Kushagra/gBaaddy). **Don't re-derive these dead ends.**
202   - **Court-masking ? multi-court fix** (see above) ? it's a **precision** tool for off-court persons and
203   one-sided feeds, a *different* lever; measure its lift, don't assume it.
204   - **Net-crossings are a strong rally signal** ? 97% of real rallies cross ?2× (median 5); single feeds ~1×.
205   Gate A leans on this.
206   - **Small-N feature discipline.** Few, scale/translation-invariant features (counts/ratios, not raw
207   coords) + a logistic regression, not a net ? raw coords memorize this court/camera.
208   - **Calibration, not representation, is the small-N wall** ? learned probabilities separate (high AUC) but
209   the operating point doesn't transfer with few training videos. More distinct matches is the highest-ROI
210   unblock.
211   - **Gate A already existed, just wasn't wired to production** ? the cheapest precision win is plumbing, not
212   new capability. Always check what the eval drivers already do that the live path doesn't.
213

8. assistant (2026-06-14T08:27:53.195Z)

I now have the empirical backbone confirmed with real numbers. The key load-bearing facts are verified: n=6 corpus, nothing significant in the leaderboard, learned model at 0.423-0.583 (not "very high"), the "Calibration not representation is the small-N wall" lesson, and the multi-court ship target of 0.66 that even Gemini struggles with. I have everything I need to write the strongest counter-case. Let me also quickly confirm the "designed for this is single-tenant by design" claim and the on-device deferral.

9. user (2026-06-14T08:27:54.940Z)

```
docs\EXPERIMENT_HARNESS.md:209:**Single-tenant / local-first** (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment registry + the
docs\NEXT_STEPS.md:127:  PLATFORM slice = async job model (single-tenant).
docs\PLATFORM_ARCHITECTURE.md:226:that first, single-tenant, then generalize.
docs\PLATFORM_ARCHITECTURE.md:243:    Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
docs\PLATFORM_ARCHITECTURE.md:474:| **P0** | **Async job model, single-tenant** (DEFERRED #16) | `jobs` table via `user_version` ladder; worker; 202+status endpoints |
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:19:- **On-device is LATER** (a model-compression goal so power users save upload cost/hassle) ? definitely on the
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:170:next platform slice remains the async job model (single-tenant); only the M0.1?M0.4 foundation is safe to start.*
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:30:- First implementation slice = **async job model (DEFERRED #16)**, single-tenant ? **pending explicit go-ahead** (no implementation started without approval).
```

10. assistant (2026-06-14T08:29:45.530Z)

All load-bearing claims are verified against the repo. The counter-case is grounded in real numbers. Here is the frank assessment.

The case AGAINST personalization-first: a frank assessment

Bottom line up front. Personalization-first is an *intellectually seductive* pivot that fits the tools the owner already built ? but it inverts the project's own validated strategy (cloud + B2B academies + generalization), bets the entire value proposition on the one statistical regime the team has repeatedly proven it cannot trust (small-N per-corpus eval), and rebrands a *generalization* metric ("good in <5 videos") as the proof of a *personalization* thesis. The strongest reading is that the harness makes per-user **tuning** cheap, which is real ? but the owner is mistaking "the bones support a special case" for "this is the right business." On the current evidence the safer, larger, and more defensible bet is generalization-first, with per-user tuning as a thin feature on top, not the strategy.

I'll take the requested points one by one, then state the conditions under which generalization-first wins.

(a) TAM / economics ? personalization SHRINKS the market and kills the only flywheel

The strategy docs already concluded the opposite of this pivot, for documented reasons, and nothing in the pivot reverses those reasons:

- **The validated beachhead is B2B academies, ranked #1 precisely because individual amateurs are "low WTP, high CAC/churn."** Personalization-first re-centers on the individual the research deliberately deprioritized. The pivot owes an argument for why per-user value *reverses* a low-WTP verdict, and "your model is tuned to you" is not self-evidently worth more to a hobbyist than "good highlights now." It is asserted, not shown.

- **The addressable user gets strictly narrower, not wider.** The cloud product gates on one constraint (static full-court camera). Personalization-first stacks two more on top: *AND has accumulated 10?15 of their own sessions AND owns a capable local GPU*. The repo's own dev container has no GPU/torch/cv2 ? the same constraint the **end user** would now inherit. That triple-gate (capture + corpus + hardware) is a smaller intersection than the cloud product's single gate. You are choosing a narrower funnel and calling it focus.

- **It trades a compounding moat for a non-compounding one.** The strategy docs name the moat as *"the consented dataset + eval harness"* ? a cross-user asset that makes the product better for the *next* buyer as it grows. A per-user-overfit model has **no network effect**: user #1,000 gets no benefit from users #1?999. Personalization-first is therefore a *tool* business (linear, per-seat, capped) masquerading as a *platform* business (super-linear, flywheel). For a solo/Bangalore-based founder, deliberately abandoning the only super-linear growth mechanism you have is the expensive choice.

- **Pricing power is capped from both sides.** Below: free app tiers and BadPro+ pay-per-use. Above: SwingVision's ~$180/yr software-only anchor. "Continually self-optimizes" is a benefit a hobbyist *cannot evaluate at purchase time* ? it's a faith claim, not a demoable feature. You can't charge a premium for an invisible delta.

Verdict: personalization-first shrinks TAM (narrower gate), removes the flywheel (no cross-user compounding), and doesn't raise the price ceiling. That is three economic strikes against, for a value prop the team's own market research already rated low.

(b) "Wins hold anecdotally on other videos but weaker" ? this is a TRAP dressed as a feature

This hand-wave is the load-bearing weakness of the whole pivot, and it's worse than it sounds for three independent reasons.

1. The repo *engineered away* the very specialization the pivot wants to sell. The features were deliberately built as scale/translation-invariant counts and ratios "because raw coords memorize this court/camera" (`fusion_features.py`, and the standing lesson "raw coords memorize this court/camera"). So the architecture *caps* how deeply a model can overfit to one user's court ? which means the per-user upside *beyond cue ON/OFF selection* is small by design. The pivot's differentiator ("deeply tuned to THEIR rallies") is partly impossible in the current feature space, and making it possible means **deliberately reintroducing the overfitting the team spent effort preventing.** You'd be tearing out a safety rail to sell the danger it was protecting against.

2. Per-user ablation optimizes for the past corpus, not the next video ? and can disable a *structural* guard. Concrete failure mode the research surfaced and I'll sharpen: Gate-A (net-crossings ? 2) is the **held-shuttle false-positive killer** ? the owner's *own observed bug* (a player flicking the shuttle between points counted as a rally). Its job is structural, not statistical. If a given user's first 10 videos happen not to contain warm-up/feed sequences, the harness measures "no lift" and the personalization logic dutifully turns the guard **OFF for that user**. Then they record one session full of between-point flicks and the model has no defense ? a regression the per-user system *caused* and has no baseline to catch. "Optimize for what this user has shown me" is exactly wrong for guards whose value is in the cases the user *hasn't* shown you yet.

3. "Weaker on others" destroys your ability to test. A model intentionally tuned to one user has **no shared golden baseline to regress against**. The entire no-regression discipline (`run_regression.py` + `regression_baseline.json`, "rollback = flip a flag") assumes a shared baseline. A deliberately-overfit per-user model would need a *per-user* no-regression harness that doesn't exist ? so a "batteries upgrade" or a per-user retrain can silently degrade a paying power-user with **no automated catch.** This directly contradicts the owner's load-bearing "no regression, ever" requirement. Personalization doesn't just lack the guard; it makes the guard structurally impossible.

Verdict: per-user overfitting is a trap. It's partly impossible (invariant features), partly dangerous (disabling structural guards on absence-of-evidence), and partly untestable (no per-user regression baseline). "Weaker on others" isn't a tolerable side effect ? it's the symptom of all three problems.

(c) Eval noise ? per-user A/B is the n=6 problem, but WORSE, and the team already watched it fail

This is, frankly, the single most damning point, and it's not theoretical ? it's in the repo's own data.

- **The team built `eval_stats.py` (bootstrap CI + paired sign test) *because* a bare LOVO mean lies at n=6.** The standing lesson is explicit: "Sign test needs n ? ~n=10/9-wins for p<0.05." A hobbyist at 10?15 videos is *at or below* that floor on a good day; a fresh user at n=1?4 is below where `leave_one_video_out` is even *defined* (it errors at <2 videos).

- **The leaderboard I just read shows NOTHING is significant at n=6.** The two recorded fusion experiments are a **2/6 loss** (p=0.6875, ?F1 CI [?0.165, +0.162]) and a **4/6 non-significant** (p=0.6875, ?F1 CI [?0.054, +0.203] ? crosses zero). The Gen-0 head's own ship gate refuses to ship (sign test 5/6, p=0.109). **Every single comparison the team has run at this N is statistically inconclusive.** Personalization-first proposes to make automated, per-user, default-flipping decisions ("turn motion OFF for you") *in exactly this regime, per user, with even fewer folds.*

- **It already collapsed concretely.** The first real A/B's learned variant scored **F1 = 0.000** on the 6-rally video ? it predicted zero rallies, a pure operating-point artifact at small N. A per-user system on a hobbyist's single short session would hit this same cliff. The team's fix (tune-threshold + calibrate inside the LOVO loop) raised n=6 mean F1 0.307 ? 0.423 ? but threshold-tuning on a tiny train fold is *itself* a small-N overfitting source, and that fix is unvalidated beyond n=6.

- **A homogeneous corpus cuts both ways.** Yes, one user's videos are lower-variance (same court/camera/lighting) ? but they also give **fewer independent folds.** Lower variance might converge faster; fewer independent matches might *never* reach significance. Which effect dominates is **unmeasured** (the research flags re-running `ab_n6_lovo.py` within-user as the highest-value experiment ? meaning the pivot's core statistical premise is currently *unconfirmed*). You'd be betting the product on an empirical question nobody has answered.

Verdict: the pivot proposes auto-promoting per-user config in precisely the regime the team's own honesty tooling exists to *refuse* promotion. Even with the honest gate, most per-user CIs will straddle zero, so the system mostly does nothing ? and when it does act, it risks locking in noise as a per-user default. "Personalization" at n?15 is, on the current evidence, indistinguishable from "fitting noise per user."

(d) Compute / maintenance ? the burden multiplies, the privacy promise is deferred

- **The heavyweight personalization (VLM fine-tune) is infeasible on a hobbyist machine ? and even inference is unmeasured.** The owned-model plan is explicit: Qwen3-VL fine-tune needs *rented* 24?48 GB A100/L40S; the owner's 2070 Super "CANNOT QLoRA a 7B video VLM"; even on-device 4B *inference* latency at 1080p/60fps is an open, unmeasured risk. So per-user VLM personalization either (i) requires cloud bursts ? which **re-introduces the DPA/consent/no-train-on-minors/cross-border machinery the local-first pivot was supposed to eliminate** ? or (ii) is restricted to the pro archetype with serious hardware. The "your data never leaves your machine" headline is only true at the **on-device milestone, which the roadmap explicitly defers** ("On-device is LATER... do not over-index on it now"). Near-term, "personalization" means per-user video + per-user labels *sitting in the cloud being trained on* ? the exact privacy surface the pivot advertises escaping. **The privacy thesis writes a check only the deferred milestone can cash.**

- **N per-user fine-tunes = N copies of every governance problem, at the edge where auditing is hardest.** The base model's pretraining provenance is already "unauditable for every open VLM incl. Qwen3-VL" and needs explicit owner risk-acceptance. Personalization *multiplies* the derivatives that inherit that debt: instead of one model to document and risk-accept, there are N per-user fine-tunes, each needing minors-exclusion + opt-in honored, each living at the edge. Plus GDPR/DPDP **unlearning**: "delete = drop the adapter" works for the per-user adapter, but a baseline that was "batteries-upgraded" on consented cohort data **cannot un-bake one user without a full retrain** ? the unlearning story for the shared baseline is undefined.

- **None of the platform plumbing exists, and it cuts against the stated guardrail.** There is no per-user model store, no per-user versioning, no per-user benchmark runner, no per-user regression baseline, no user/tenant dimension anywhere (`user_id` appears only as a *future* comment). The harness is explicitly "single-tenant / local-first... NOT MLflow/W&B/a service." Federated coefficient averaging (the only way the correction loop feeds the shared baseline without a video lake) has **no DP noise, no secure aggregation, no minimum-cohort floor** ? and with a 6-dim coefficient vector over few users, a contributed update is close to a per-user fingerprint (membership-inference surface). The "size of baseline.json" guardrail and "continually self-optimizes for N users" are in direct tension.

Verdict: the cheap layer (thresholds + a kilobyte logistic) is genuinely cheap; everything that would make personalization *feel deep* (VLM fine-tune, on-device privacy, per-user stores, federated upgrades, unlearning) is either infeasible-now, deferred, or unbuilt ? and each one re-opens a governance question the pivot claimed to close.

(e) Cold-start for shot types ? the cold-start chasm, but with class imbalance on top

Shot taxonomy is the part of the pivot's product story ("deep analysis of YOUR serves/smashes/drives") that is *most* exciting and *least* supported:

- **It's entirely greenfield.** A whole-codebase grep finds `shots_count` only as a CSV header in a test fixture; there is **zero** shot-type or direction-reversal detection code. There is no "battery" to self-optimize ? it must be built from scratch, unlike rallies where a heuristic + Gen-0 head already exist.

- **Class imbalance, not just small N, is the killer.** Rallies are ~single-class with 10?15 positives per short video. Shot taxonomy is 6?7 classes *dominated* by serves/clears/drives, so the product-interesting minority classes (smash, drop, net) may be **near-absent** in a hobbyist's 10?15 videos. "Heavily optimized by 10?15 videos" is already shaky for rallies; for minority shot classes per user it's fantasy.

- **The correction loop doesn't produce shot labels.** Today's correction surface is rally-level confirm/reject. Confirming "this is a rally" yields *no* serve/smash/drop label. Capturing them needs a **new, finer, higher-effort per-shot correction UI** ? exactly the friction the casual hobbyist won't absorb. The "correction loop as per-user shot-label source" is unvalidated for the archetype that's supposed to be the funnel.

- **The empirical reality check is sobering.** Even RALLY detection isn't "heavily optimized by 15 videos": shuttle-only LOVO F1 **0.307**, learned-with-threshold **0.423**, person cue gives **negative** lift at n=6. The ship target on the hard multi-court case is 0.66 ? which *Gemini itself* only just reaches. Shot taxonomy strictly **multiplies** a data hunger rally detection hasn't satisfied. And the labels are *noisier* (ROORA v8 flags shot-type as needing its own inter-annotator-agreement check), so per-user self-labels are lower-quality precisely where data is scarcest.

Verdict: the deep-analysis layer that sells the pivot pulls strategy *back toward generalization-first* ? you need a strong cross-user vendor-labeled shot baseline before any per-user refinement is meaningful, which is the opposite of personalization-first for this layer. Promising per-user shot tuning to hobbyists is, on the evidence, unfalsifiable for that archetype.

(f) "The harness was designed for this" ? a genuine *mechanism* fit, but post-hoc as a *strategy* claim

This deserves a precise verdict because it's doing a lot of rhetorical work for the owner.

True at the mechanism layer: `compare(videos, control, variant)` is corpus-relative ? "this user's corpus" is literally the list you pass in. Per-cue flags, Gate-A/B thresholds, the pinnable classifier path, the honest small-N stats, the replayable per-video feature JSONs ("detect once, decide forever") ? all of these *do* compose into per-user cue selection with little new code. The owner is **right** that the bones support per-user *tuning*.

Overstated as "designed for this":
- **Three load-bearing pieces are stubs or absent.** (1) The learned per-user model has **no serving path** ? `LogisticRegression.load` is never called in `backend/pipeline`; production reads only the Gate thresholds. So "optimize a per-user model" currently produces an artifact the live pipeline *cannot consume*. (2) The correction-loop?labels write path **does not exist** ? `human_label` is never SET in production (only a test writes it via raw SQL); there's no correction/reject API in `main.py`. The most load-bearing assumption of the pivot is the *least* built. (3) There is **no user/tenant dimension** ? single-corpus by design, explicitly "NOT a service."
- **No design doc, test, or comment names personalization as a target.** Every doc frames the harness as "add cues without regressing the SHIPPED baseline" ? a *generalization-first* goal. The same primitives *happen* to compose into per-user optimization. "The bones support it" is true; "it was designed for it" is retrospective fit.

Why this matters: "designed for this" is being used to imply low cost ("we already built it"). But the harness supports per-user **tuning** (config the served segmenter reads) far better than per-user **learning** (a model nothing in production loads). If the pivot leans on "deep analysis via a learned model," the **serving gap is the first wall**, and the correction-loop write path + per-user store are the next two. The architecture asymmetry is the tell: the cheap, near-ready thing (cue ON/OFF) is *not* the differentiated thing; the differentiated thing (a model deeply tuned to YOU) is the *unbuilt* thing. The owner is conflating "the easy 20% is done" with "the hard 80% is implied."

Verdict: a genuine advantage for a *modest* feature (per-user cue/threshold selection), a post-hoc rationalization for the *strategic* claim ("the roadmap was designed for personalization-first"). The harness was designed to protect a generalization-first baseline; it incidentally enables per-user tuning.

When generalization-first wins instead (the conditions)

Generalization-first is the better bet under conditions that, on the current evidence, largely **already hold**:

1. **If statistical power is the binding constraint** ? which the repo's own data screams (nothing significant at n=6; "more distinct matches is the highest-ROI unblock"). Generalization-first pools all matches into one corpus, maximizing N and folds; personalization-first *fragments* the data into per-user piles, each below the significance floor. When you're data-starved, you pool ? you don't shard.

2. **If the buyer is the academy (aligned recorder=payer, B2B, higher WTP)** rather than the corpus-rich-individual ? which the validated PMF research already concluded. A generic model that works on *any* academy's first upload beats a model that needs each academy-student to accumulate 15 personal videos.

3. **If "good on video #1" is the make-or-break metric** vs instant-cloud incumbents (SwingVision/BadPro+ deliver on the first clip). The cold-start chasm is fatal for the median user; a strong generic baseline *is* the day-1 product, and the only currently-validated win ? **Gate-A, label-free, +0.074 F1, 6/6, p=0.031** ? needs zero per-user labels. That's a generalization win, available now.

4. **If the moat is meant to be the dataset/flywheel** (the strategy docs' explicit thesis). Generalization-first keeps consented data compounding into one ever-better shared model; personalization-first severs that loop.

5. **If the "<5 videos to superior quality" promise is to be provable at all** ? because that is a **held-out-USER (generalization) metric**. Improving it *is* the generalization-first objective wearing a personalization hat. You cannot honestly certify the baseline on biased per-user signal; you need an independent vendor-graded golden set. So even the pivot's own headline KPI is a generalization KPI.

6. **If governance simplicity matters** for a solo founder ? one centrally-auditable model (provenance, minors-exclusion, unlearning, opt-in) beats N edge-resident derivatives each

carrying the same debt.

The honest synthesis the owner should hear: the right architecture is almost certainly **generalization-first baseline + a thin per-user tuning layer on top** ? exactly what the harness already nearly supports (per-user cue ON/OFF + threshold selection, gated behind the honest ?F1 sign-test with a hard minimum-N floor, falling back to the shipped label-free Gate-A/B below that floor). That captures the *real* and *cheap* part of the personalization idea (transparency feature: "we tested the motion cue on YOUR videos and it didn't help, so it's off for you") **without** betting the company on per-user overfitting, small-N auto-promotion, on-device VLM fine-tuning, or a shattered data flywheel.

Personalization-first as *the strategy* fails because it inverts the validated market (away from academies), abandons the only compounding moat, auto-promotes in the one eval regime the team proved it can't trust, defers its own privacy headline to an unbuilt milestone, and rebrands a generalization metric as personalization proof. Personalization as *a feature* ? built on a generalization-first baseline ? is the version worth keeping.

Files that ground this assessment (all verified this session, absolute paths):
- `C:\Users\avidu\Projects\badminton-highlight-indexer\eval_baselines\experiment_leaderboard.json` ? the two recorded fusion A/Bs, both non-significant (p=0.6875), CIs cross zero. The empirical heart of point (c).
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md` ? n=6 numbers (shuttle 0.307 / learned 0.423 / person negative), the multi-court 0.66 ship target, Gate-A promotion (+0.074, 6/6, p=0.031), and the standing lesson "Calibration, not representation, is the small-N wall ? more distinct matches is the highest-ROI unblock."
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md` (lines ~19, ~170) ? "On-device is LATER... do not over-index on it now"; single-tenant async-job slice is the committed next platform step.
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EXPERIMENT_HARNESS.md` (§9, ~line 209) ? "Single-tenant / local-first... NOT MLflow/W&B/a service" (the "designed for generalization, not per-user" evidence for point (f)).